The Impact of LLMs on Sponsored Search: Evidence from Google's BERT *

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Abstract

Large language models (LLMs) are transforming the way digital platforms engage and interact with text. Such changes have the potential to significantly impact downstream platform users. We investigate the impact of LLM-based query interpretation systems on search engine sponsored search markets, focusing on cost-per-click (CPC) and the number of advertisers bidding for ad space (competition). We begin with a theoretical model in which a search engine utilizes a language model to interpret consumer search queries and generate information to estimate advertiser relevancy scores for sponsored ad space. We find that LLM-based interpretation can (1) increase bidder eligibility across queries and simultaneously (2) yield heterogeneous CPC shifts: higher CPC for low-context queries (i.e., short queries) and declining CPC for high-context queries. The decline in CPC is driven by LLM's ability to understand context, resulting in increased dispersion of relevancy scores and the prioritization of highly relevant advertisers in the auction. We empirically test these predictions using monthly data on the number of bidders and CPC for 12,000 queries and exploiting Google's 2019 rollout of Bidirectional Coding Representation from Transformers (BERT) as a natural experiment. Consistent with our model, we find that the number of bidders competing for ad space increases across queries, CPC rises for short, context-poor queries, and falls for longer, context-rich queries. Our results demonstrate how LLM-based interpretation systems can impact sponsored search markets.

1 Introduction

In June 2017, researchers at Google Brain introduced the transformer model architecture (Vaswani et al., 2017). Notable for their ability to dynamically accommodate contextual language information through the "self-attention" mechanism, transformers underpin modern large language models (LLMs). These models have significantly improved programmatic text interpretation (encoding) and generation (decoding) capabilities and have the potential to reshape digital platform markets.

In online search, search engines rely on natural language processing (NLP) algorithms to infer user intent from queries, converting raw text into relevance signals. These signals help the search engine identify and rank relevant advertisers for sponsored search auction opportunities. The use of LLMs to interpret search queries can alter how a search engine processes queries and subsequently matches and ranks advertisers for advertising opportunities. Any changes in this market are consequential for both search engines and advertisers, as sponsored search continues to be a dominant advertising channel for firms to reach consumers and remains the primary source of revenue for many search engines. In 2023 alone, US search advertising revenue reached nearly $90 billion.[1]

In this paper, we ask: how does the implementation of LLM-based interpretation systems on search engine platforms impact sponsored search markets? We address this question by theoretically and empirically studying how Google's October 2019 implementation of Bidirectional Encoding Representations from Transformers (BERT) impacts short-term query-level cost-per-click (CPC) and the number of bidders present in query advertising auctions (i.e., market size).

The answer to this question is not trivial and depends on how LLM information affects advertiser relevancy scores. In sponsored search, query interpretation algorithms provide search intent signals to the platform (Google) to help estimate advertiser relevancy scores

[1] See: `https://www.iab.com/news/2023-u-s-digital-advertising-industry-hits-new-record-according-to-iabs-annual-internet-advertising-revenue-report/`

(i.e., click-through-rate (CTR) likelihood). These scores are used to measure the match quality (i.e., relevance) between an advertiser and a search query, which impacts both auction eligibility and the final paid prices through an advertiser's ad rank score.[2] An advertiser's final ad rank score depends on both the bid they submit and the search engine's estimated relevancy score. Higher relevancy scores will boost an advertiser's ad rank, increasing their likelihood of being allocated to an auction and winning an ad slot. Higher relevancy scores also reward advertisers with lower final prices, with the expectation that they have a higher expected CTR.

On the one hand, an LLM-based interpretation model could help a search engine better interpret search queries and identify more relevant advertisers, leading to more bidders with sufficiently high relevancy scores being allocated into auctions. Suppose the search engine is cautious about allocating irrelevant advertisers. In that case, more informative signals about the search intent of a given query may help it identify relevant bidders more often, leading to potentially thicker markets and higher prices. On the other hand, new information may help search engines segment query markets and remove irrelevant advertisers from auctions, leading to smaller average market size.

The effect on prices (CPC) is ambiguous due to potential changes in the number of bidders competing for a search query and advertiser relevancy scores. Holding all else equal, a higher (lower) number of bidders allocated to the auction will drive prices up (down). If, over time, advertisers learn that the match quality is better and increase their bids, then prices may also increase. However, if winning advertiser relevancy scores increase or relevancy score dispersion between advertisers increases, then final prices may decline due to sponsored search mechanics rewarding highly relevant advertisers with lower prices.

We begin addressing this question by developing a stylized theoretical auction model.[3]

[2]See `https://support.google.com/google-ads/answer/1722122?hl=en` for a discussion of Google ad rank and use of relevancy scores.

[3]The focus on the short run is due to data limitations that do not allow us to study and test long-run effects. Specifically, in March 2020, the COVID-19 pandemic started and likely affected consumer, advertiser, and platform behavior in ways that could directly affect our empirical estimates (e.g., more people spending more time at home and online).

Our model considers a platform (e.g., Google) that faces a two-sided matching problem: match relevant advertisers to incoming consumer search queries. The platform utilizes an algorithm to learn about a query's search intent and subsequently uses the algorithm's output to identify and rank relevant advertisers based on their relevancy scores. These scores impact auction eligibility and final paid prices. Selected advertisers then submit bids and compete in an auction. Advertisers are ranked with ad ranks, which depends on the submitted bids and the platform's estimated relevance scores. Advertisements are sold in a second-price pay-per-click (PPC) auction.

The model analysis focuses on the comparative statics of this partial equilibrium model to understand how the introduction of a new interpretation algorithm may impact relevancy scores and our primary market outcomes (CPC and number of bidders). The model yields several interesting results.

First, our model predicts that all queries will generally see thicker, not thinner, markets. This translates to, on average, more bidders per auction. The cost of showing irrelevant advertisements drives this result. Search engines worry about the negative externalities associated with showing irrelevant advertisements, leading to search engines not allocating every advertiser to every sponsored search auction. More informative signals generated by an LLM increase the platform's confidence that it understands a query's focus, which leads to more relevant bidders being identified, driving the average number of bidders per query up.

Second, the model predicts that significant improvements in contextual understanding within a query moderates shifts in CPC. For context-rich search queries (e.g., longer queries such as "shoes with no laces"), our model predicts that short-run prices may fall despite an increase in the average number of bidders competing for advertising space. In these auctions, contextual understanding allows the platform to estimate more precise relevancy scores. This increases the overall relevance dispersion among competing advertisers and leads to highly relevant advertisers (i.e., those with a high expected click-through rate, or CTR) being prioritized within the auction. In contrast, less contextually relevant advertisers, who are still related, are de-prioritized.

Consider the query “shoes with no laces”. Without understanding the context “with no laces”, the platform cannot further differentiate “shoe” bidders, such as Crocs or Sorel, within the auction. The platform recognizes that both bidders are relevant, but cannot quantify how relevant. By understanding the context within the query, the platform can differentiate advertisers, prioritizing Crocs and deprioritizing Sorel due to Crocs being more contextually relevant (they primarily sell shoes without laces). It still allows Sorel to compete because they relate to “shoes”, but their relevancy ranking is lowered because the context suggests they’re less relevant within the “shoes” space. This separation in relevancy scores can result in lower final prices, even when more bidders compete in the auction.

For queries with limited contextual information (e.g., short queries such as “shoes”), our model predicts that prices increase with the expected increase in the number of bidders. The inherent lack of context limits the platform’s ability to further differentiate bidders within these auctions, resulting in limited relevancy score dispersion. While Google can identify that Sorel might be related to “shoes”, it lacks additional contextual information within the query to determine whether Sorel, Crocs, or Puma are more or less relevant.

We rely on Google’s October 2019 introduction of BERT to test these theoretical predictions. In October 2018, Google researchers used the transformer architecture to build BERT, one of the first LLMs. BERT was notable for its ability to understand context and for its dramatic improvement in predictive tasks over previous state-of-the-art models (Devlin et al., 2018). In October 2019, Google introduced BERT as one of its main interpretation algorithms to parse and understand user-generated search queries.[4]

To study BERT’s effect, we collect monthly average CPC and Competition Score (CS, a

[4]See `https://blog.google/products/search/search-language-understanding-bert/` for the official BERT announcement. See `https://searchengineland.com/welcome-bert-google-artificial-intelligence-for-understanding-search-queries-323976` for industry announcement. See `https://twitter.com/searchliaison/status/1204152378292867074` for the announcement of international BERT, stating that US English BERT was introduced in October 2019.

measure of the number of bidders participating in an auction) from SEMRush—a company that tracks search query performance—for a sample of roughly 12,000 search queries over two years (2018-2020).

To estimate the effect of BERT on CS and CPC, we provide two separate identification strategies. The first strategy relies on variation in search query vector representations to identify which queries are more or less affected by the introduction of BERT. In doing so, we compare differences in CS and CPC before and after the introduction of BERT across queries that are more or less likely to be impacted by BERT. This strategy leverages variation across queries to identify the effect of BERT.[5] The second strategy exploits the panel nature of our data and employs a difference-in-differences (DD) approach akin to those discussed in Eichenbaum et al. (2020), Bollinger et al. (2022), and Liaukonyte et al. (2022). Specifically, we compare changes in CPC and CS before and after the introduction of BERT, using a baseline of changes over the same months and queries from the previous year. This strategy exploits variation within queries and across time to identify BERT's effect. Across both identification strategies we find consistent results.

We first test for changes in market size via changes to CS. We find that on average, CS increases by 0.1-0.3%, implying that more bidders are competing in ad auctions. We then group queries based on their length to test for market changes based on the degree of contextual information present.[6] Consistent with the theoretical model, while CS increases by roughly 01-0.4% for short and long queries, CPC increases for short queries by 1.5% and declines by -2.5% for longer queries. Despite more bidders competing for long query ad space, prices are lower than they were before the introduction of BERT. The shifts in CPC correlate with the amount of context present within the query and is consistent with our theoretical predictions.

To summarize, we find that the implementation of LLM-based interpretation algorithms

[5]Several papers relied on this type of identification strategy in the past, including Zervas et al. (2017); Farronato and Fradkin (2022); Card and Krueger (2000); Duflo (2001); Akerman et al. (2015).

[6]In general, the amount of contextual information present within queries increases with query length.

can have a significant and immediate impact on sponsored search markets. The new information generated by LLMs can help search engines estimate more precise relevancy scores, which, surprisingly, can lead to lower CPC despite more bidders competing for advertising space. This is because understanding context enables the platform to better differentiate bidders within auctions. However, context-poor queries do not benefit from such effects and will see short-term prices rise. For advertisers interested in sponsored search, this means that LLM-based interpretation systems will reward targeting highly relevant, longer queries, while also making shorter queries more competitive and costly. As an advertiser, expanding ad reach to less relevant queries will become increasingly more difficult due to improved precision in relevancy scores leading to lower ranks. To overcome these changes, advertisers will need to submit higher bids to maintain their market share.

Combined, our theoretical model and empirical findings help advertisers understand how LLMs used to interpret search queries immediately alter sponsored search markets and impact downstream market size and prices. In doing so, we also highlight why understanding context in language is economically significant. For academics, our empirical findings contribute to the growing literature on the economic impact of AI and LLMs on markets.

2 Related Work

Our paper relates to the growing literature on the economics of AI and LLMs, sponsored search, information disclosure in auction markets, and targeted advertising.

Economic impact of AI and LLMs Recent advancements in LLM technology have spurred academic interest in understanding the potential economic effects of these models. More recently, the development of generative LLMs has motivated researchers to study how these models impact labor markets (Eloundou et al., 2023; Zarifhonarvar, 2023), information markets such as Stack Overflow and Reddit (Burtch et al., 2023), and marketing practices (Kushwaha and Kar, 2021; Reisenbichler et al., 2022; Goli and Singh, 2024). Recently, Padilla et al. (2025) studied how user adoption of generative LLMs (e.g., ChatGPT) influences consumer search. They find that generative LLMs substitute other online search tools when consumers are looking for objective information or answers to clearly specified queries. However, online searches for LLM-adopting users spike at the beginning and then stay flat for 20 weeks. After 20 weeks, they observe a decline in searches. These results suggest that consumers do adjust their search behavior when interacting with new search systems, but that such adjustments occur slowly and take several months.

Importantly, all of the NLP models described in these papers stem from the transformers architecture (Vaswani et al., 2017). We contribute to this growing literature by studying how predictive LLMs used to interpret consumer search queries for traditional search engines impact sponsored search auction markets. Our theoretical model also highlights the economic significance of understanding context in language, a technological novelty associated with the transformer architecture.

Sponsored search Sponsored search remains a prevalent advertising channel for businesses. It also remains an active area of research in both marketing and economics (Edelman et al., 2007; Ghose and Yang, 2009; Yang and Ghose, 2010; Rutz and Bucklin, 2011; Berman and Katona, 2013; Blake et al., 2015; Edelman and Lai, 2016; Simonov et al., 2018; Cowgill and Dorobantu, 2020; Sahni and Zhang, 2024). Motivated by the auction structure of sponsored search, Edelman et al. (2007) and Varian (2007) study equilibrium bidding strategies in generalized second price auctions. Empirical work has also analyzed ad effectiveness (Ghose and Yang, 2009; Blake et al., 2015), sponsored and organic complementarities (Yang and Ghose, 2010), keyword spillovers (Rutz and Bucklin, 2011), and advertising competition Simonov et al. (2018). One stream of research focuses on the downstream consequences of search engine platform design decisions, including the impact of a search engine's services on click behavior (Edelman and Lai, 2016), result page features (Gleason et al., 2023), and the interaction between search engine optimization (SEO) and sponsored links (Berman and

Katona, 2013). We contribute to this literature by empirically and theoretically studying how improved search engine interpretation algorithms affect sponsored search markets.

Information disclosure in auctions Research on information disclosure in auction markets has primarily focused on settings where sellers can endogenously hide or reveal information to buyers. Related to our setting, Cowgill and Dorobantu (2020) empirically studies how disclosing new information to advertisers in the sponsored search market affects prices, profits, and CTR. The authors find that information disclosure generally leads to thinner markets and lower prices but higher overall profits due to improved CTR. These market adjustments are due to advertisers improving their self-selection into preferred markets, leading to better query-advertiser matching.

Tadelis and Zettelmeyer (2015) also examines how information disclosure influences buyer participation in auction opportunities. The authors find that information disclosure in the automobile resale market can help quality-differentiated buyers self-select into preferred auction opportunities, leading to higher market clearance rates and higher profits across all quality types.

Our setting differs from existing papers. In our context, buyers do not receive any information to help with market selection or bid adjustment. Instead, the seller (Google) privately receives the interpretation algorithm information and uses it to pick and rank buyers for auction opportunities. As we will see, this market structure leads to new empirical findings that differ from previous work.

Targeted advertising Our paper also relates to a stream of literature on matching and targeting technology improvements in advertising markets (Bergemann and Bonatti, 2011; Ada et al., 2022). Amaldoss et al. (2016) studies how improvements to sponsored search broad match technology affect market entry and seller profits. The authors find that better broad match *bidding* algorithms can lower market entry costs and induce greater auction participation, leading to higher prices.

Empirical and theoretical work has also documented the benefits and market effects of better-targeted advertising technology. Chandra (2009) empirically studies the newspaper market and finds that better targeting can lead to higher prices due to improved advertiser-audience alignment. Athey and Gans (2010) theoretically studies how targeting technology can lead to an increase in the supply of advertising opportunities, potentially putting downward pressure on prices. Similarly, Bergemann and Bonatti (2011) studies how improved targeting technologies given to advertisers can improve matching efficiency, increase advertiser bids, but lower prices and market size. We contribute to this literature by examining how more informative search query intent signals provided to a platform, rather than advertisers, influence sponsored search market size and prices.

3 Empirical Context

Sponsored search advertising When a consumer types a query into a search engine (e.g., "how to cook chicken"), a search engine must interpret the query and decide what domains (URLs) to present on the Search Engine Results Page (SERP). At the top of the SERP, search engines may offer sponsored search links. These links are bid for in a real-time auction before the SERP loads on the consumer's web browser.

Sponsored search auctions follow a pay-per-click (PPC) model. Under PPC, advertisers pay only when a consumer clicks on an ad. Because search engines allow multiple sponsored search links to appear at the top of result pages, most rely on a Generalized Second-Price Auction (GSPA) to sell ad space and rank buyers (Edelman et al., 2007; Varian, 2007). Under a GSPA, advertisers are ranked by bids, and the top N winners for N ad slots show advertisements. If an advertisement receives a click, the advertiser pays the minimum price needed to out-price the next highest bidder (usually by \$0.01).

In practice, advertisers are ranked with Ad Rank scores. Ad Rank scores depend on many factors, including the advertiser's bid and their platform's estimated relevancy score. The relevancy score measures how related an advertiser is to a particular query. Higher scores indicate that the advertiser is more relevant and is predicted to receive a click. Ad relevancy matters because it impacts auction eligibility and CPC; higher relevancy scores increase the likelihood of auction participation and lower final paid prices. [7]

The organic rank results are below the sponsored links on the SERP. These are links that the search engine deems relevant to the given search query. Unlike sponsored links, advertisers do not purchase organic links. Instead, the search engine uses an internal ranking system to decide positions. A firm can appear in both sponsored and organic search engine results. For example, even if Nike bids (and wins) the top ad slot in the sponsored search position for the query "women's running shoes," it can still appear in the organic rank results.

BERT Bidirectional Encoder Representations from Transformers (BERT) is a large-scale neural network-based language model developed by Google in 2018. It is a pre-trained model capable of understanding the context and nuances of natural language text, making it a powerful tool for a wide range of NLP tasks (Devlin et al., 2018).

BERT's novelty stems from its ability to understand the context in language objects (Devlin et al., 2018; Tenney et al., 2019b; Rogers et al., 2021). Effectively capturing contextual information was a significant challenge for previous state-of-the-art NLP algorithms such as Word2Vec (Mikolov et al., 2013) and was considered a significant breakthrough in NLP research. With contextual knowledge, BERT can better understand language semantics, for example, when "bank" indicates a financial institution or land alongside a river. It can also lead to a better understanding of complex syntactic language structures, such as adjectives and adverbs, as well as non-linear linguistic relationships. To capture this information, BERT generates a vector for each word based on the other words surrounding the focal word using the transformer-based self-attention mechanism (Vaswani et al., 2017). As such, the same word can have a different vector representation depending on its surrounding words.

BERT can impact the interpretation of queries with and without explicit contextual

[7] See `https://support.google.com/google-ads/answer/1722122?hl=en`.

information. For longer queries with explicit contextual information, BERT will improve Google's ability to understand how the words within the query interact and connect. BERT can also change the interpretation of context-poor short queries. Consider the simple query "bank". During its training process, BERT will learn that "bank" can potentially relate to financial institutions or rivers. Even though the query itself does not explicitly contain context, the underlying training process enables BERT to learn these implicit associations and incorporate that information into the word's vector representation, thereby altering the overall interpretation.

Google deployed BERT in October 2019. In its release notes,[8] Google stated that "by applying BERT models to both ranking and featured snippets in Search, we're able to do a much better job helping you find useful information". In other words, Google uses BERT to identify which websites are related to a specific search query (likely by computing similarity scores using vector representations of queries and websites). While Google does not explicitly mention in its press release how this could affect search ads, discussions with current and former employees suggest that signals generated from models like BERT would be readily accessible to the sponsored search team for ranking advertisers.

4 Modeling the Market

To aid with hypothesis generation and to better understand how LLMs can immediately impact sponsored search markets, we consider a stylized model with a two-sided matching market where a platform uses text to match potentially relevant advertisers to incoming query advertising opportunities. To facilitate matching, the platform relies on an interpretation algorithm to generate signals about the search intent of a search query. The platform then uses these signals to estimate advertiser relevancy scores.[9] Using these scores, the platform selects relevant bidders to compete in the auction. Selected bidders submit bids and are

[8] https://blog.google/products/search/search-language-understanding-bert/

[9] See https://support.google.com/google-ads/answer/1722122?hl=en for discussion of Google ad rank and use of relevancy scores.

ranked within the auction based on their ad rank, which is a combination of their bid and the platform's estimated relevancy score. The winning advertiser (i.e., the one with the highest ad rank) is displayed, and a click is received based on the true match quality between the winning advertiser and the query. The winning advertiser pays only if a click is received.

The model has two key components. First, we assume that queries are multidimensional. More specifically, a query is defined by its topic (i.e., the focus) and its context (i.e., linguistic modifier information). Each dimension is distinct, and interpretation algorithms generate signals across each dimension. This multidimensional data structure is motivated by extensive literature in multiple disciplines, including linguistics, computer science, and neuroscience, which commonly models language objects as multidimensional (Mnih and Hinton, 2008; Jäger and Rogers, 2012; Miyagawa et al., 2013; Coopmans et al., 2023). Second, we assume the platform aims to maximize its own profit and exclusively receives interpretation signals. Advertisers do not observe the interpretation signals generated by the platform's language model, nor do they observe their relevancy scores before the auction. This is consistent with the fact that advertisers on Google do not observe their relevancy scores at bid time or know the exact information that is being used to estimate their scores.

A query in our model is defined by two components: the topic (focus) of the query and its contextual modifier information. In NLP, topic modeling is a prevalent area of research. Even before BERT, NLP algorithms like Latent Dirichlet Allocation (LDA), Word2Vec (W2V), or Doc2Vec (D2V) organized textual documents into latent "topics". Ultimately, topic modeling focuses on categorizing and clustering similar documents (Blei et al., 2003; Mikolov et al., 2013; Karvelis et al., 2018). The notion that language objects, such as queries, can be classified into topic buckets implies that there is differentiation across topic categories.

Contextual information is distinct from topical information and plays a fundamentally different role within queries and language objects. Context conveys additional meaning that differentiates queries *within* a topic category. Although both "shoes with no laces" and "shoes with laces" topically relate to "shoes", they have different types of contextual information that imply different meanings. The context dimension leads to differentiation *within* a given topic category.

Prior to the introduction of transformers, effectively modeling and interpreting contextual information at scale and in a realistic manner was a significant challenge. The transformer architecture substantially improved contextual interpretation capabilities, representing a substantial breakthrough in NLP research.

Yet, not all queries explicitly contain context, limiting the potential for differentiation within a given topic category. The short query "shoes" topically relates to "shoes with no laces" and "shoes with laces" but lacks additional contextual information to differentiate its search intent within the topic category. The inherent difference in information conveyed within the search query distinguishes shorter queries from longer queries, limiting the information that can be learned using an NLP algorithm.

To summarize, queries are multidimensional. They can be bucketed and differentiated across topics and further differentiated within topics based on context, but only when context is present. We capture this structure in our model by defining a query as having a topic dimension that represents the focus of the query and a context dimension that defines the modifier information present within the query. A topic and a context dimension will also define an advertiser's text-based advertisement. The CTR function for a query will depend on the importance of the topic and context alignment between the winning advertiser and the query. A platform's interpretation algorithm generates distinct topic and context signals.

While not novel to our setting, we assume that advertisers are horizontally differentiated (e.g., Nike and Geico are relevant to different queries). Additionally, we assume the platform pays a negative externality cost when irrelevant advertising is shown. Irrelevant ads can annoy consumers, increase consumer search costs, or harm the reputation of the platform (Broder et al., 2009; Berman and Katona, 2013; Lu et al., 2017). We now describe our model.

4.1 Setup

Advertiser An advertiser A_i is defined by a topic $\theta_i \in \{0, 1\}$ and a context $z_i \in \{0, 1\}$. We assume four bidders are in the market, one for each combination of topic and context ($< 0, 0 >$, $< 0, 1 >$, $< 1, 0 >$, and $< 1, 1 >$). Advertisers maintain a private value $v_i \sim U[0, 1]$ for clicks.[10]

Query A query Q is defined by a topic $t \in \{0, 1\}$, a context $q \in \{0, 1\}$, and a topic importance parameter $B_t \in [\frac{1}{2}, 1]$. θ_i and z_i directly map to t and q types for queries, respectively. There are four possible query types ($< 0, 0 >$, $< 0, 1 >$, $< 1, 0 >$, and $< 1, 1 >$). t captures the underlying topic of the query (e.g., is it about "shoes" or "insurance") while q captures contextual modifier information. Each query has an equally likely probability of being drawn ($\frac{1}{4}$).

Denote the advertiser who wins the advertising opportunity for query Q by A_w, its type by θ_w, and its context by z_w. The CTR for Q depends on the match quality between the winning advertiser and the query. B_t is the relative importance of topic alignment, compared to context alignment, to the query's CTR. We model query CTR in Equation 1.

$$CTR(A, Q) = B_t I_{\{\theta=t\}} + (1 - B_t) I_{\{z=q\}}, \tag{1}$$

While a topic and a context define each query in our model, B_t determines the importance of alignment across each dimension. Context-rich queries (i.e., longer queries) will have smaller B_t values, meaning context alignment is important, while context-poor queries (i.e., shorter queries) will have larger B_t values. This captures the intuition that for a short query like "shoes", context alignment does not significantly impact the query's CTR because the context isn't meaningfully present. In contrast, for a long query like "shoes with no laces",

[10]We assume valuations $v_i \sim U[0, 1]$ because advertisers do not observe match quality before the auction and therefore cannot change short-term valuations. A two-period model aimed at studying long-term effects could relax this assumption and allow advertisers to learn over time what the new match quality is, leading to changes in valuations that depend on the topic and context dimensions.

context alignment does matter to the query's CTR.

Platform Our model considers a platform (Google) that sells query ad space to advertisers. Ad space for a query gets sold via a second-price auction (SPA). Like Google's matching process, the platform endogenously selects which advertisers participate based on relevancy constraints. The advertising space is sold on a pay-per-click (PPC) basis, meaning the winning advertiser only pays when their ad is clicked.

The platform's profit function is:

$$\pi(\hat{A}, Q) = I_{\{Click\}} \frac{\tilde{Adr}}{r_w} - I_{\{\theta_w \neq t\}} C, \tag{2}$$

Where $\frac{\tilde{Adr}}{r_w}$ is the second highest Ad Rank in the auction scaled by the relevancy score of the winner bidder, $I_{\{Click\}}$ indicates a click, $I_{\{\theta_w \neq t\}}$ indicates when there is a mismatch between the query topic and winning advertiser topic, and C is the negative externality cost associated with displaying an irrelevant advertisement to the consumer (i.e., topically irrelevant ad). $\frac{\tilde{Adr}}{r_w}$ is structured such that winning advertisers pay the minimum bid price needed to win the position (Berman and Katona, 2013; Amaldoss et al., 2015).

Denote the set of chosen advertisers for an auction opportunity by $\hat{A}$. Before selection, we assume the platform observes advertiser types (θ and z). However, the platform needs to use an algorithm to interpret and identify the context and topic of a query. When a query Q arrives, the platform uses an algorithm $a \in \mathbb{R}$ to interpret it. For a query Q with topic t and context q, an algorithm a takes Q as input and outputs a signal $\hat{Q}$ with components $\hat{t}$ and $\hat{q}$ to the platform.

For a query Q and algorithm a, $Pr(\hat{t} = t) = \gamma_t(a)$ and $Pr(\hat{q} = q) = \gamma_q(a)$. The platform learns the true value of t with probability $\gamma_t(a)$, and it receives an inconclusive signal with probability $1 - \gamma_t(a)$. Similarly, the platform learns the true value of the context q with probability $\gamma_q(a)$ and it receives an inconclusive signal with probability $1 - \gamma_q(a)$. We assume the inconclusive signal is equivalent to the platform's prior: $Pr(t = 1) = Pr(q = 1) = \frac{1}{2}$.

Therefore, $\hat{t} \in \{0, \frac{1}{2}, 1\}$ and $\hat{q} \in \{0, \frac{1}{2}, 1\}$. We also assume that $\frac{\gamma_t}{\partial a} > 0$ and $\frac{\gamma_q}{\partial a} > 0$, i.e., a better algorithm always improves the probability of learning the true value of t and q. Finally, we assume advertisers and the platform know $\gamma_t(a)$ and $\gamma_q(a)$.[11]

After receiving the estimates $\hat{t}$ and $\hat{q}$, the platform picks which advertisers, if any, to allocate to the auction opportunity to maximize expected profits. Conditional on being assigned to an auction, it is a weakly dominant strategy for advertisers to bid their valuations v_i (the proof is in Appendix A). We model the expected CPC for a query with advertiser set $\hat{A}$ by $E[CPC(\hat{A})] = E[CTR] * E[\frac{\tilde{Adr}}{r_w}]$.

Ad ranks determine the order of bidders. We model ad rank for an advertiser i as $Adr_i = b_i r_i(\hat{Q})$, where b_i is the submitted bid and $r_i(\hat{Q})$ is the relevancy score advertiser i receives from the platform given the information set $\hat{Q}$. The relevancy score is modeled as the expected CTR for advertiser A_i given signal $\hat{Q}$: $r_i(\hat{Q}) = Pr(\theta_i = t|\hat{t})B_t + Pr(z_i = q|\hat{q})(1-B_t)$. We assume that the platform knows the CTR coefficient B_t.

Timing The order of the game is as follows. A query Q is randomly drawn by Nature, producing unobservable components q and t. The platform uses algorithm a to generate signals $\hat{q}$ and $\hat{t}$. The platform then estimates relevancy scores for each advertiser and selects a subset of auction participants. If the seller runs an auction, chosen advertisers submit bids, the winning advertiser displays an ad, the consumer decides whether to click, given the type alignment, and the platform receives a profit. An algorithm a affects profits, CPC, the average number of query bidders, and CTR by changing the seller's q and t signals.

4.2 Results

Our results will focus on auction outcomes at the query level and show how improving the algorithm a can impact CPC and the Number of Bidders (NOBs) competing for advertising opportunities in the sponsored search auction market. Interestingly, we will show that under

[11]This assumption stems from the fact that Google announces the algorithm update and releases press articles that describe how new algorithms work.

reasonable assumptions, query-level CPC may decline despite an increase in the average number of bidders. We now walk through how this can occur in this partial equilibrium model. Formal proofs are in Appendix A.

The average auction market CPC and the number of bidders depend on the cost of showing topically irrelevant advertisements (C) and the underlying importance of topic alignment (B_t) to a query's CTR. If C is small, the platform is willing to allocate potentially irrelevant advertisers to auctions. Even if it shows an irrelevant advertisement, the cost of doing so is low, and its overall expected profit is higher because the auctions have more bidders competing. If C is large, the platform is unwilling to allocate potentially irrelevant advertisers to auctions. Here, the platform errs on the side of caution when allocating bidders because it doesn't want to incur the cost C. When C is moderate, the average CPC and number of bidders largely depend on the underlying importance of topic alignment (B_t). In combination, C and B_t create six distinct regions, leading to different platform selection strategies and ultimately different expected NOBs and CPCs. Figure 1 visualizes these regions based on B_t and C,[12] and Lemma 1 presents the expected NOBs and CPC by region.

Lemma 1. *The average CPC and NOBs for a query Q depend on C and B_t. Let* $J = \frac{40-60B_t+45B_t^2-16B_t^3}{30(2-B_t)^2}$, $X = \frac{9+18B_t-3B_t^2+16B_t^3}{30(1+B_t)^2}$, $Z = J + X - \frac{B(3-B)}{6}$, $P = \frac{-5+20B_t-25B_t^2+15B_t^3-3B_t^4}{6B_t}$, *and* $L = X + J - P$. *Table 1 presents average NOBs and CPC for a query by region.*

Region 1 in Lemma 1 describes the average CPC and NOB when C is large, while Regions 5 and 6 dictate average CPC and NOB when C is small. Regions 2 through 4 describe average query CPC and NOB when C is moderate. When $C > \frac{8}{5}$, all queries live in Region 1.

In Table 1, notice that when C is low (Regions 5 and 6), NOB declines with larger topic or context signals (γ_q and γ_t). When C is low, the platform errs on the side of allocating all advertisers, even if some of those advertisers are potentially irrelevant. When search intent signals improve, the platform removes the irrelevant bidders. Regions 5 and 6 capture the intuition that better interpretation algorithms can segment markets.

[12] For the exact thresholds of C that define the regions, see Appendix A.5.

Figure 1: Regions that determine expected CPC and NOBs based on C and B_t. C is the cost the platform incurs for showing a topically irrelevant ad and B_t is the importance of topic alignment for a query's CTR.

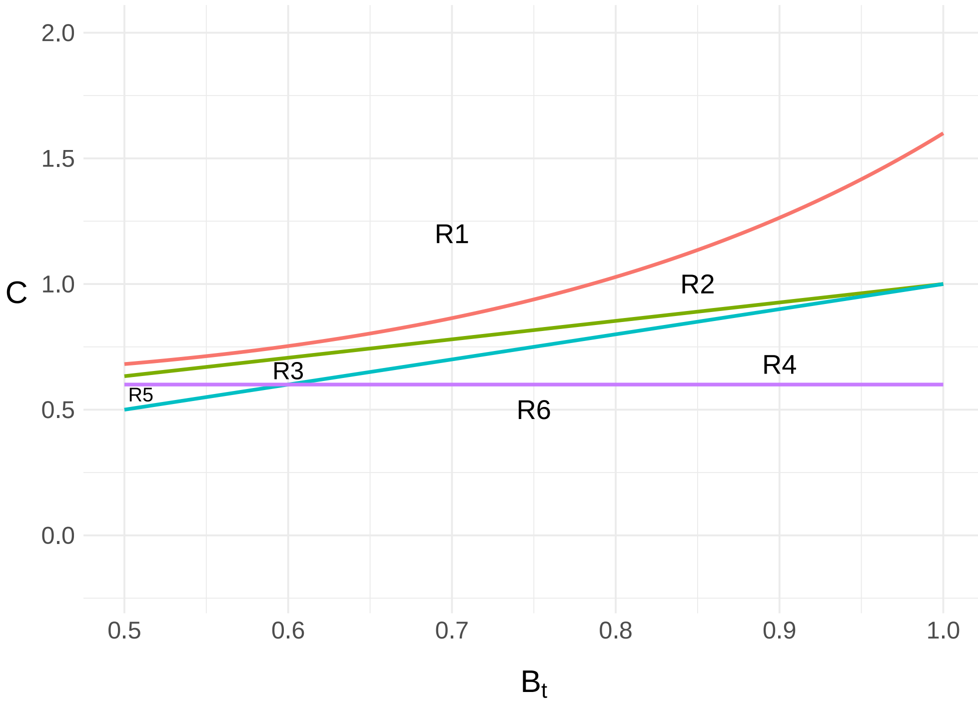


Table 1: Expected Number of Bidders and CPC across model regions.

Regions	$E[NOBs]$	$E[CPC]$
Market Expansion (Higher C)		
Region 1 (R1)	$2\gamma_t$	$\frac{\gamma_t}{6}[1 + B_t - \gamma_q(B_t - 1)^2]$
Both (Moderate C)		
Region 2 (R2)	$2\gamma_t + 4\gamma_q - 4\gamma_t\gamma_q$	$\gamma_t\frac{1+B_t}{6} + J\gamma_q - \gamma_t\gamma_q(J + \frac{(B_t-1)^2}{6})$
Region 3 (R3)	$4(\gamma_t + \gamma_q) - 6\gamma_t\gamma_q$	$J\gamma_q + X\gamma_t - Z\gamma_t\gamma_q$
Region 4 (R4)	$4(\gamma_t + \gamma_q) - 5\gamma_t\gamma_q$	$X\gamma_t + J\gamma_q - L\gamma_t\gamma_q$
Market Segmentation (Lower C)		
Region 5 (R5)	$4 - 2\gamma_t\gamma_q$	$(J - \frac{3}{10})\gamma_q + (X - \frac{3}{10})\gamma_t - (Z - \frac{3}{10})\gamma_t\gamma_q$
Region 6 (R6)	$4 - \gamma_t\gamma_q$	$(X - \frac{3}{10})\gamma_t + (J - \frac{3}{10})\gamma_q - (L - \frac{3}{10})\gamma_t\gamma_q$

When C is high (Region 1), the platform is highly conservative in allowing advertisers into the auction. Here, the platform prefers to run an auction only when it has informative topical signals; otherwise, no auction is run. This "relevancy constraint" restricts advertiser allocation to auctions. When topic signals improve, the number of bidders actually increases, as the platform can do a better job of avoiding C and allocating topically relevant bidders. Here, the platform errs on the side of not allocating under uncertainty, and signal improvements overcome this uncertainty, leading to market expansion outcomes rather than market segmentation outcomes.

For brevity, we assume C is large ($C > \frac{8}{5}$) and focus on Region 1 for the rest of the discussion. We refer the reader to Appendix A.4 for a discussion of the other regions.[13]

Market size changes In Region 1, the platform is extremely conservative about allocating advertisers to auctions because the cost of showing topically irrelevant advertisements is high. This leads to an average number of bidders for a query Q of $2\gamma_t(a)$. Intuitively, the platform will only allocate advertisers when they know they're in the right topical "ballpark". If the platform knows an incoming query is about "shoes", it is happy to allocate "shoe" advertisers. But, if the platform is unsure whether a query is about "shoes" or "insurance", it's not willing to allocate any bidders for fear of incorrectly showing an "insurance" advertiser on a "shoe" query, or vice versa.

When the platform improves the quality of its interpretation algorithm (increasing the value of a), it can generate more informative topic signals, leading to relevant advertisers being allocated more often. This leads to Proposition 1.

Proposition 1 (Platform Selection of Advertisers)**.** *Under sufficiently high cost C ($C \geq \frac{8}{5}$), the average number of bidders for all query ad space increases with a ($2\frac{\partial}{\partial a}\gamma_t(a) > 0$).*

Proposition 1 contrasts results from Bergemann and Bonatti (2011) and Cowgill and Dorobantu (2020) by finding that improvements to targeting technologies can increase, not decrease, market size. As discussed, this is a byproduct of the platform's market selection process, which aims to maximize profits and avoid C.

[13] Anecdotal evidence from talking with former employees at Google suggests that Google is extremely cautious about bidder selection and worries about platform quality and reputation, suggesting that C is indeed large.

CPC changes Average CPC for a query in Region 1 is $\frac{\gamma_t}{6}[1+B_t-\gamma_q(B_t-1)^2]$ and depends on both γ_t and γ_q. Improvements to topic signals help the platform get relevant advertisers into the auction, putting upward pressure on average CPC. At the same time, context signals help the platform further differentiate relevant bidders. Doing so increases relevancy score dispersion and can put downward pressure on CPC.

To build intuition for the effects of context, consider the case when there are two topically relevant bidders competing for ad space. The platform initially only knows that they are topically relevant, so they each have the same relevance score, r. Valuations are both drawn from $U[0, 1]$, so they have the same ad rank distribution $U[0, r]$. Competition is intense because their ad rank distributions are identical, and each has a 50% chance of winning.

Now, consider when the platform identifies the context of the query and further differentiates the two bidders. Based on the query context, one bidder is highly relevant and receives a high relevancy score $r < r_H$, while the other is less relevant and receives a lower relevancy score $r_L < r$. The bidder with r_H has an ad rank distribution of $U[0, r_h]$ and the low-relevance bidder has a distribution $U[0, r_L]$ with $r_L < r_H$. This differentiation causes the expected second-highest ad rank score to fall and the expected CTR to increase. If the degree of separation caused by identifying context is substantial, the decline in expected second-highest ad rank scores can result in a decline in CPC.

When the platform improves its algorithm a, both its topic and context signals change. This leads us to Proposition 2.

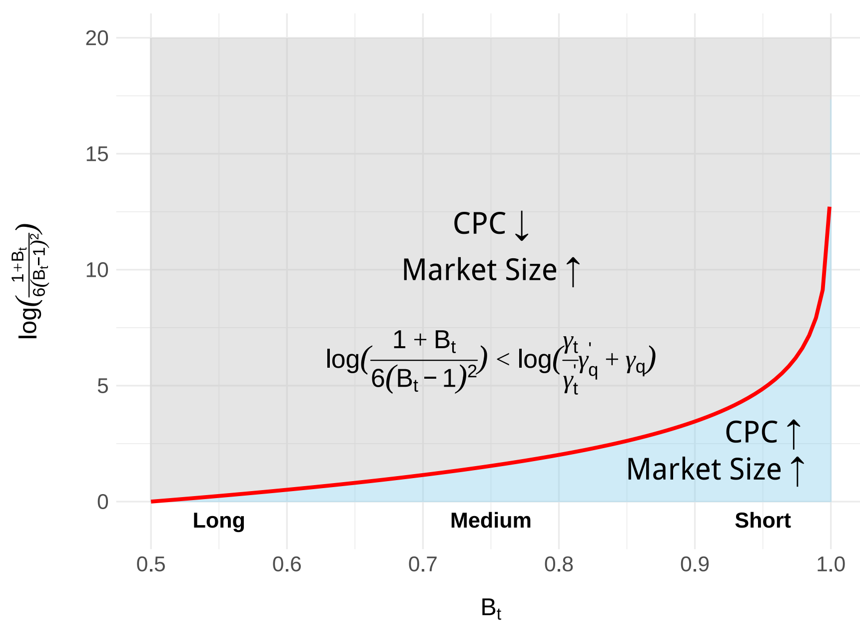
Proposition 2 (Average CPC). *Define γ_q' and γ_t' as the partial derivatives of each parameter with respect to a. Average CPC for a query in Region 1 will decline if $\frac{1+B_t}{6(B_t-1)^2} < \frac{\gamma_t}{\gamma_t'}\gamma_q' + \gamma_q$, increase if $\frac{1+B_t}{6(B_t-1)^2} > \frac{\gamma_t}{\gamma_t'}\gamma_q' + \gamma_q$, and see no change at equality.*

After introducing a new algorithm, Proposition 2 says that CPC is more likely to decline when context alignment matters to a query's CTR function (e.g., long queries with low B_t) and context signal improvements are large (i.e., shifting from a non-transformer model to a transformer-based LLM model). Because context is not prevalent in short queries (large

B_t), further bidder differentiation is neither meaningful nor possible, making it unlikely to see a declining CPC.

In Figure 2, we present the threshold value $log(\frac{1+B_t}{6(B_t-1)^2})$ by B_t.[14] The gray area of the figure indicates when CPC declines (i.e., $log(\frac{1+B_t}{6(B_t-1)^2}) < log(\frac{\gamma_q^{'}}{\gamma_t^{'}}\gamma_t + \gamma_q))$ despite an average increase in the number of bidders. The blue region indicates where CPC would increase with the increase in the number of bidders. Finally, the line represents equality. As B_t decreases, the potential for lower CPC increases. In other words, longer queries have a greater potential to experience a decline in CPC after a new algorithm is introduced, due to the prevalence of context.

Figure 2: Potential market changes in Region 1. Query length buckets are indicated on the X-axis.



Summary and predictions To summarize, our model suggests that enhancements to a platform's interpretation algorithm enable the platform to estimate more precise relevancy scores for advertisers. When externality costs associated with showing irrelevant advertise-

[14]We log scale the inequality for visualization and indicate what regions of B_t short, medium, and long query lengths would live.

ments are large, a new algorithm leads to thicker markets for all queries due to topic signal improvements helping the platform identify relevant advertisers more often. For context-rich queries, when context signal improvements are substantial, platforms may experience lower CPCs due to significant improvements in bidder differentiation. For context-poor queries, there is a lack of contextual information to differentiate bidders further, making it difficult for CPC to decline.

Based on our model's results, we make the following predictions about BERT's impact to Google's sponsored search market. First, incremental improvements in topic signals will lead to expanded markets, including more advertisers for existing auctions and more queries in competitive auctions. This is driven by BERT's ability to better understand the focus of a query, and this will occur for all queries. Second, we expect BERT to substantially improve Google's context signals. This will predominantly benefit existing long query auctions (those with low B_t) and may lead to lower CPCs despite an average increase in the number of bidders. Existing short query auctions (high B_t) will not benefit from context signal improvements and will see a higher CPC associated with the increase in the number of bidders.

5 Empirical Analysis

5.1 Data

Sampling search queries To empirically test our model predictions, we collect data from SEMRush, a leading provider of sponsored search rank and keyword data. We rely on a survey that we administered to Amazon Mturkers to generate a sample of queries for the analyses presented in this paper. After selecting 32 different topics (see Appendix B for the list of topics), we asked survey respondents the following question: "Please write a search query related to the topic of '**Topic**' that you would search for on Google". 100 Amazon Mturk participants took the survey and each participant was asked about five randomly selected categories, giving us 500 responses and roughly 16 responses per topic category.

Not all MTurk submissions were queries. To focus on submissions that were search queries, we removed 210 irrelevant answers. Irrelevant responses included answers where participants typed in specific URLs, website content, descriptions of how to type up a search query, and quotes from Google "Help Pages" describing how to search.

Using the remaining 290 sampled queries, we turned to the broad match database at SEMRush.[15] For a given query, SEMRush returns a set of similar queries for which it has data for the current month.[16] We limited our match to queries with an average search volume of at least 500 from 2021-2022, generating a dataset of roughly $120,000$ queries related to the original 290 queries. However, many of the queries contained explicit terms (e.g., adult content such as pornographic search queries) unsuitable for analysis (advertisers generally don't target adult content queries). To filter these out, we generated a list of adult content terms and fuzzy-matched each query with the list of terms. We removed the query from our data if there was significant similarity between the query and any of the terms. After completing this process, we retained roughly $40,000$ queries.

CPC and Competition Score SEMRush provides two key outcomes relevant to answering our research question: CPC and Competition Score. CPC measures the average price advertisers pay for an ad click for a query. Competition Score is a proprietary normalized score (between 0 and 1) that measures the relative number of advertisers bidding for ad space for the given query.[17] It is worth noting that SEMRush collects data for many queries, even if they have little to no advertising, meaning CPC and Competition Score can be zero.

We collected historical monthly information about CPC and Competition Score for each of these queries from January 2018 to February 2020. We then filtered out queries that

[15] See: `https://www.semrush.com/features/keyword-research/`

[16] For example, for the query "shoes", the database will show monthly data for "shoes" as well as similar queries, such as "running shoes", "women's shoes", "best shoes for hiking", etc.

[17] From discussions with SEMRush, Competition Score is linearly comparable and can be viewed similarly to the Google Trends information commonly used in research to measure search volume. While we can't estimate the precise number of bidders, we can estimate relative changes.

lacked persistent historical information (more than a year of missing data) and those with limited historical search volumes (less than 100 average searches/month during the time frame). This left us with $11,949$ unique queries and $284,146$ monthly observations from January 2018 to February 2020. In the top part of Table 2, we present overall summary statistics for our dependent variables, Competition Score and CPC, across all queries during this time window.

Table 2: Summary statistics.

	Mean	St. Dev.	Min	Max	Median
	All queries				
CPC	1.25	3.343	0.00	584.73	0.54
Competition score	0.244	0.356	0.00	1.00	0.05
	Short queries				
CPC	0.911	1.918	0.00	64.110	0.400
Competition Score	0.248	0.364	0.00	1.00	0.050
	Medium queries				
CPC	1.339	3.947	0.00	584.730	0.610
Competition Score	0.278	0.378	0.00	1.00	0.060
	Long queries				
CPC	1.533	3.078	0.00	83.860	0.720
Competition Score	0.126	0.216	0.00	1.00	0.030

Table 2 shows that, in the aggregate, 1) many queries are not competitive (low median and mean Competition Score) and 2) CPC is often relatively low, but the maximum price can be extremely high. We also break down queries by word length, an important moderator we focus on in our analysis. We categorize queries into three groups for ease of presentation. A short query has two or fewer words, a medium query has three to five words, and a long query has six or more words. We present summary statistics by query length in the bottom part of Table 2. The table shows that as query length increases, demand generally decreases while CPC increases. These results generally align with the trend that, as query length increases, the queries become more niche and are harder to match but also potentially more valuable (higher CTR) to advertisers.

To empirically validate CS and CPC, we collect SEMRush and Google Keyword Planner (GKP) tool data from July 2021 to February 2022 for 10,000 queries in our dataset.[18] From GKP we have two measures of competition: (1) "Competition", which is a measure of the number of bidders competing in the query's auction using three values: low, mid, and high; and (2) "Competition (index value)" which measures the total number of ad slots filled divided by the total number of ad slots available on a scale from 0 to 100. The correlation between these two measures is 0.98. We then compute the Pearson correlation between the SEMRush's Competition Score and these two measures. We find that the SEMRush's Competition Score has a correlation of 0.65 with Google's "Competition" score and a correlation of 0.87 with the "Competition (index value)" [19], suggesting SEMRush data is reasonably correlated with data obtained directly from Google.

To validate CPC, we use the only CPC-related measure from GKP: the maximum submitted bid. Although average CPC and maximum bid are different measures, they should reasonably correlate if SEMRush data is reliable. The Spearman correlation between SEMRush CPC and Google's maximum submitted bid is 0.57, suggesting a high level of agreement.

6 Primary Identification Strategy

To study how BERT impacts CS and CPC, we implement two difference-in-differences (DD) identification strategies that rely on different underlying assumptions. Our primary identification strategy leverages variation in linguistic properties to distinguish queries that are more or less affected by the introduction of BERT. In doing so, we compare queries more likely to be affected by BERT with queries less likely to be affected by BERT (first difference) before and after the introduction of BERT (second difference). In Appendix D, we

[18]Unfortunately, GKP data goes back in time to at most three years, and we did not collect data in time to replicate some of the analysis reported in this paper with GKP data.

[19]We encode Google's Competition measure as low (0), medium (0.5), and high (1) to calculate this correlation.

present our secondary identification strategy that exploits within-query variation. We find consistent results across both strategies.

6.1 Defining Query Metrics

To operationalize our first identification strategy, we define two linguistic metrics to identify a query's treatment intensity.

The Computer Science literature has documented that BERT better understands complex syntactic language structures and semantic relationships between words and sentences (Devlin et al., 2018; Lin et al., 2019; Tenney et al., 2019a,b; Rogers et al., 2021). A complex query will benefit from BERT's ability to capture syntactic information, while all queries, including those with simple syntactic structures, will experience changes to semantic relationships and understanding.

Motivated by these observations, we create two measurement variables, Linguistic Complexity and Cosine, to capture query syntactic complexity and semantics. Under the assumption that the interpretation of these linguistic properties changed with the introduction of BERT, we can identify variation in our dependent variables caused by BERT's implementation through BERT's interaction with these measurements.

Linguistic Complexity BERT can better understand syntactic language structures (Lin et al., 2019; Tenney et al., 2019a). Therefore, BERT's introduction should change how Google handles syntactic language. Examples of syntactic information include query syntax tree structures (Lin et al., 2019), parts of speech (Tenney et al., 2019a), and sentence dependency features Tenney et al. (2019a); Liu et al. (2019). Hewitt and Manning (2019) finds that syntax trees, i.e., hierarchical characterizations of grammatical language structures, are embedded in BERT vector spaces and Tenney et al. (2019b) finds that Parts of Speech (POS) tags are also present in BERT vector spaces. These findings help us identify the information BERT will interact with and drive the design of our first measurement: Linguistic Complexity (LC).

For each query, we measure the depth of the syntax tree and count the unique number of Parts of Speech (POS). LC is defined as a dummy variable and takes on a value of 1 if either a query's syntax tree depth is greater than the median depth in our dataset (median = 2) or the unique POS count is greater than the median count in our dataset (median = 2). Otherwise, it's 0. LC captures the syntactic complexity of the query.

Cosine BERT understands semantics and how words and concepts *relate* to each other (Tenney et al., 2019b). It knows that socks relate to shoes, banks can relate to bodies of water or financial institutions, and mice can relate to computers or rodents. This semantic knowledge improves Google's ability to interpret and categorize search queries, even those without explicit context. This latter observation is critical, as changes to query relations likely impact Google's matching process and identification of relevant advertisers. Understanding that a query relates to more (fewer) queries will likely lead to more (fewer) advertisers being deemed relevant.These observations motivate the design of our second linguistic property: Cosine.

At a high level, we measure semantic changes (Cosine) using the difference in the number of queries related to a given query before and after BERT.

To define Cosine, we must first identify the primary interpretation algorithm used by Google before BERT (RankBrain). Google's preceding algorithm, RankBrain, is likely Doc2Vec (D2V) or Word2Vec (W2V), the former being a more generalized form of W2V.[20]

For a query i, we generate a vector representation with BERT and D2V models. We then calculate the cosine similarity score between query i and all other queries in the dataset *within* a given model vector space. This generates two $N \times N$ cosine similarity matrices, one for the BERT vector space and one for the D2V vector space, where N is the number of queries in our dataset and matrix element i, j is the cosine similarity score between query i and query j.

[20]Google filed patents: `https://patents.google.com/patent/US9740680B1/en`. See also: `https://opensource.googleblog.com/2013/08/learning-meaning-behind-words.html` for technology before RankBrain that looks similar to Word2Vec, and the researchers for both Word2Vec and Doc2Vec were employed at Google.

We use these matrices to measure changes in query semantic relationships across D2V and BERT vector spaces. Specifically, we measure the set of queries that pass "relatedness" thresholds *within* vector spaces and then compare differences in these sets *across* vector spaces. We first calculate each cosine matrix's 25th, 50th, and 75th percentile cosine scores. These are our "relatedness" thresholds.[21] Then, for each query i, we identify the set of other queries with a cosine similarity score greater than or equal to the chosen threshold (i.e., 25, 50, or 75) in the respective vector spaces. For example, if the 75th threshold in the D2V space is 0.5 and the query "shoes" has a cosine similarity score of 0.8 with the query "socks" and 0.2 for the query "hat", then the "socks" query will be in the relevant set for the D2V space but not the "hat" query. If, in the BERT space, the 75th threshold is 0.8 and the cosine score between "shoes" and "hats" is 0.9, then "hat" would appear in the relevancy set for "shoes" in the BERT space. This gives us two sets of relevant queries per query, one for each vector space.

Then, we calculate the differences between the relevance sets across BERT and D2V for a given query. We define "added" queries as those that did not appear in the D2V relevancy set but did appear in the BERT set. Alternatively, we define "removed" queries as those that appeared in the D2V set but did not appear in the BERT set. For a given query i, we sum up the total number of "added" and "removed" queries and take the ratio between "added" and "removed". We define Cosine as this ratio.

A ratio value of one indicates that the same number of queries were added as removed. Ratio values less than one indicate that more queries were removed than added, indicating that BERT believes the query is more specific and related to fewer other queries. Finally, ratio values greater than 1 tell us that more queries were added than removed, suggesting that BERT believes the query is related to more queries. Because the ratio of added to removed queries is skewed to the right (median = 1.094, mean = 175.605), we use the log of

[21] We vary the thresholds to generalize the measurement. We focus on thresholds because we reason that Google likely has some cut-off requirement determining whether an advertiser is relevant enough to a given query.

1 + Cosine in our model.

It is worth noting that we use this process to generate Cosine because the D2V and BERT vector spaces and their cosine scores are not directly comparable without making strong assumptions about what the dimensions of each model's vector space represent. D2V and BERT capture potentially different sets of information, making vector comparison and projection methods infeasible. Additionally, the scales of these vector spaces are potentially different, meaning we cannot directly compare query cosine scores across algorithms. Therefore, to effectively measure query semantic changes, we generate standardized measurements *within* vector spaces such that they can then be compared *across* vector spaces.[22]

Average query metrics Table 3 and 4 present average Cosine and LC measures, respectively, for all queries ("All") and by keyword length. Table 4 also presents the proportion of queries with POS counts and syntax tree depths greater than the median. (A value of 1 means above the median for the particular measurement). We identify several patterns. First, LC measures correlate with query length due to the presence of within-query contextual information, which inherently makes them longer. Second, short queries have, on average, a large Cosine score, implying that BERT recognizes they relate to more queries, while long queries have a low Cosine score. The average long query ratio between "added" and "removed" is 0.78, indicating that BERT finds these queries more specific than D2V.

Table 3: Average Log(Cosine+1).

Keyword Length	75th Ratio	50th Ratio	25th Ratio
All	1.761	1.324	1.049
Short	3.107	2.348	1.645
Medium	1.393	1.043	0.914
Long	0.650	0.481	0.451

These values are the average log(Cosine + 1), where Cosine is the ratio between added and removed sets.

[22] In Appendix C.1, we show that our results are robust to how we define Cosine and Linguistic Complexity.

Table 4: Average Linguistic Complexity.

Keyword Length	Linguistic Complexity	Tree Height	POS
All	0.613	0.566	0.400
Short	0.004	0.001	0.004
Medium	0.811	0.730	0.439
Long	1.000	0.998	0.986

6.2 Specification

Given the two linguistic metrics, we estimate the following model:

$$Y_{i,t} = \beta_1 Post_t \times LC_i + \beta_2 Post_t \times log(Cos_i) + \beta_3 Post_t \times LC_i \times log(Cos_i) + \delta_i + \gamma_t + \epsilon_{i,t}, \quad (3)$$

where $Post_t$ takes on a value of 1 for the four months post-BERT (November 2019 to February 2020), and 0 for the four months pre-BERT (July 2019 to October 2019). Our data span from July 2019 to February 2020, excluding any potential effects of COVID-19. LC_i is Language Complexity of query i and Cos_i is the the Cosine of query i. δ_i are query fixed effects and γ_t year-month fixed effects. β_1 captures the variation in $Y_{i,t}$ that is explained by the effect of LC after BERT gets introduced, β_2 captures the variation in $Y_{i,t}$ explained by the impact of changes to Cosine after the introduction of BERT, and β_3 captures any additional interaction between LC and Cos after the introduction of BERT. Because our measures are imperfect and linguistic information may overlap across measures, the interaction helps absorb residual correlation between them and mitigate collinearity in the main effects. Given that the outcomes are likely serially correlated within a query, we cluster standard errors at the query level.

Our two primary dependent variables are the log of CS and CPC. To test our model predictions and maintain consistency in the set of queries across our two identification strategies, we restrict our primary analysis to queries with existing auctions before the introduction of BERT. To do this, we retain queries where the mode CS score from July to October in 2018 and 2019 is greater than 0. This leaves us with $10,199$ queries for our primary analysis.[23]

Identification assumptions This identification strategy relies on the assumptions that: (1) outcomes for queries with high and low values of Linguistic Complexity and Cosine are comparable; (2) the linguistic properties we defined affect CPC and CS only through their interaction with the new query interpretation system (BERT); (3) BERT interacts with our linguistic measurements differently than RankBrain (the previous algorithm); and (4) there are no time-varying confounds that correlate with BERT's introduction and particular linguistic properties.[24]

To partially validate these assumptions, we perform three tests. First, we employ an event study-like model to demonstrate that pre-treatment outcomes evolved similarly for both more and less likely to be affected queries (which partially validates assumption 1). Second, we perform a placebo test in the previous year over the same months window (July 2018 to February 2019) and find insignificant effects, suggesting that the variation our measures capture is likely due to the introduction of BERT (partially validating assumptions two and three). Third, to account for potential shifts in consumer search behavior and changes to organic rankings, we show that our results hold when we control for search volume and changes to organic results (partially validating assumption four). Finally, we provide some evidence that advertisers do not seem to adjust to BERT by changing their spending.

Event study Our identification hinges on the assumption that outcomes should evolve similarly in the absence of treatment. To partially validate this assumption, we can estimate an event study where we replace $Post_t$ in Equation 3 with a month-of-year indicator that

[23]We focus on queries with existing auctions to minimize potential floor effects associated with non-competitive auctions, as these queries can only see CS and CPC increase or stay near 0. Restricting to existing auctions allows us to explore potential declines in both dependent variables. In Appendix D.1, we test for shifts in CS and CPC among non-competitive queries. Consistent with our theory model predictions, we find that both increase post-BERT.

[24]For example, changes in search behavior that are query-type specific and unrelated to BERT can be problematic. Consider the case in which wealthier consumers begin submitting search queries that are more linguistically complex because of something other than BERT. If advertisers adjust their targeting strategies and increase their bids in response to this behavior, our results will be upwardly biased.

compares the particular month to October 2019 (baseline). In Table 5 and Table 6, we present event study estimates for CS and CPC for the 75^{th} cosine percentile threshold.[25] In both tables, we see that the estimates are minimal and close to zero in the pre-treatment periods, suggesting that outcomes did evolve similarly before BERT.

Table 5: Event Study: $log(CS)$.

Month	(1) LC	(2) Cosine	(3) LC × Cosine
July	0.0001	−0.00002	−0.0001
	(0.0003)	(0.00005)	(0.0001)
August	−0.001**	−0.0001**	0.0002
	(0.0002)	(0.00004)	(0.0001)
September	−0.0002	−0.00005	0.0001
	(0.0001)	(0.00003)	(0.00005)
November	0.003*	0.0004	−0.001**
	(0.001)	(0.0003)	(0.0004)
December	0.005**	0.001*	−0.002*
	(0.002)	(0.001)	(0.001)
January	0.006**	0.001**	−0.002*
	(0.002)	(0.001)	(0.001)
February	0.002	0.002**	−0.002*
	(0.003)	(0.001)	(0.001)

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Regression includes year-month and query fixed effects. Each row observation reports the estimated coefficient for the interaction between the month-indicator and the variable reported in the column. Standard errors clustered at the query level are reported in parentheses. Data from July 2019 to February 2020. BERT is introduced in October 2019 (baseline month).

[25] We find consistent patterns across other thresholds, and present results for the 75^{th} threshold for brevity.

Table 6: Event Study: $log(CPC)$.

Month	(1) LC	(2) Cosine	(3) LC × Cosine
July	0.0003	0.0003	−0.001
	(0.002)	(0.0003)	(0.001)
August	0.002	0.0004	−0.001
	(0.002)	(0.0003)	(0.001)
September	−0.001	−0.0003	−0.0002
	(0.001)	(0.0002)	(0.0005)
November	−0.019*	−0.001	0.003
	(0.011)	(0.002)	(0.004)
December	−0.029	0.008*	−0.002
	(0.018)	(0.004)	(0.006)
January	−0.034*	0.008**	0.00002
	(0.018)	(0.004)	(0.006)
February	−0.024	0.008**	−0.002
	(0.018)	(0.004)	(0.006)

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Regression includes year-month and query fixed effects. Each row observation reports the estimated coefficient for the interaction between the month-indicator and the variable reported in the column. Standard errors clustered at the query level are reported in parentheses. Data from July 2019 to February 2020. BERT is introduced in October 2019 (baseline month).

6.3 Results

Market size results Our theoretical model predicts that BERT will enhance topic signals, enabling Google to categorize queries better. This will lead to Google identifying more relevant bidders for query auctions, thereby increasing the average number of bidders per query auction. We test this prediction by analyzing changes in CS.

In Table 7 we present Equation 3 estimates with CS as the dependent variable. $Post_t \times$

LC_i is positive and significant, suggesting that when Linguistic Complexity is above the median, CS increases by $0.33-0.4\%$. These findings suggest that when syntactic information acquisition opportunities are substantial, BERT's information increases Google's confidence in correctly interpreting the query, leading to increased market size. Second, $Post_t \times Cos_i$ is also positive and significant. The coefficient estimates suggest that a 1% increase in Cosine translates to about a 0.1% increase in CS. Finally, the interaction term between LC, Cosine, and Post is negative. Linguistically complex queries are generally more specific, and BERT learns this, leading to slower relative increases in the number of bidders.

Table 7: The effect of BERT on $log(CS)$

	(1) 75th	(2) 50th	(3) 25th
Post × Linguistic Complexity	0.0040^{**}	0.0033^{*}	0.0033^{*}
	(0.002)	(0.002)	(0.002)
Post × Log(Cosine)	0.0012^{**}	0.0009^{*}	0.0007
	(0.001)	(0.001)	(0.001)
Post × Linguistic Complexity × Log(Cosine)	-0.0015^{**}	-0.0020^{**}	-0.0030^{**}
	(0.001)	(0.001)	(0.001)
Observations	63,474	63,474	63,474
R^2	0.9786	0.9785	0.9785

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Regressions include year-month and query fixed effects. Standard errors clustered at the query level are reported in parentheses. Data from July 2019 to February 2020.

To put these results into context, we use the average keyword length values in Table 3 and Table 4 to bootstrap predicted changes to the average number of bidders for all queries and queries of varying length. We find that aggregate CS increases by $0.1-0.3\%$. When we predict average values by keyword length, we find that, across lengths, CS is increasing by about $0.1-0.4\%$ (See Table 8). The uniform increase in CS is consistent with our theoretical model predictions.

Table 8: Predicted $log(CS)$ estimates by query length.

Keyword Length	(1) 75th	(2) 50th	(3) 25th
All	0.003***	0.002***	0.001***
Short	0.004***	0.002***	0.001***
Medium	0.003***	0.002***	0.001***
Long	0.004***	0.003***	0.002***

Note: 99% confidence intervals are estimated by bootstrapping predictions (1000 iterations). ***p<0.01.

CPC results Our model predicts that CPC can increase or decrease depending on the amount of context present within the query and the relative improvements to contextual understanding. In particular, CPC should increase for queries with limited to no contextual information and potentially decrease for queries with high amounts of contextual information. Because BERT significantly improves contextual understanding, we expect it to cause CPC to decline for longer queries.

In Table 9, we present Equation 3 estimates with $log(CPC)$ as the dependent variable. We find that a 1% increase in Cosine positively increases CPC by about 0.6%. We also find that $Post_t \times LC_i$ is negative and significant, implying that for more linguistically complex queries, CPC decreases. Finally, the triple interaction is close to zero and insignificant.

We again use the average values in Tables 3 and 4 to put these results into context and to compare across query lengths. We present the estimated changes in CPC in Table 10. We find that the average CPC decreases by roughly 0.8%. When we break it down by keyword length, we find that CPC increases for short queries by roughly 1.5%, while it decreases by about 1.6% for medium queries, and by approximately 2.5% for long queries. These results are consistent with our theoretical model, which predicts that context moderates changes in CPC. For longer queries, CPC decreases because they contain more contextual information that Google can use to differentiate bidders within auctions.

Table 9: The effect of BERT on $log(CPC)$

	(1) 75th	(2) 50th	(3) 25th
Post × Linguistic Complexity	-0.0270^{*}	-0.0289^{**}	-0.0307^{**}
	(0.015)	(0.014)	(0.015)
Post × Log(Cosine)	0.0056^{*}	0.0065^{*}	0.0075
	(0.003)	(0.004)	(0.005)
Post × Linguistic Complexity × Log(Cosine)	-0.00003	0.0009	0.0006
	(0.005)	(0.006)	(0.009)
Observations	63,474	63,474	63,474
R^2	0.795	0.795	0.795

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Regressions include year-month and query fixed effects. Standard errors clustered at the query level are reported in parentheses. Data from July 2019 to February 2020.

6.4 Robustness Checks

To further reinforce the fact that it is the introduction of BERT that drives our outcomes, we perform several additional tests. First, we present a placebo test using data from the previous year, during which no known algorithm change occurred. Second, we directly address the concern that consumer behavior change (and, in turn, advertising outcomes) could occur after BERT by accounting for changes in consumer search behavior and organic results. Third, we provide descriptive evidence that advertisers do not adjust budgets in the post-BERT months, suggesting that our results are unlikely to be driven by advertisers immediately adjusting behavior. Finally, in Appendix D, we replicate our results using our alternative identification strategy.

Placebo test Our identification rests on the assumption that our observed effects are due to the interaction of linguistic variables with a new interpretation algorithm (BERT in our case). We perform a placebo test that helps us test this assumption. We estimate

Table 10: Predicted $log(CPC)$ estimates by query length.

Keyword Length	(1) 75th	(2) 50th	(3) 25th
All	-0.007^{***}	-0.008^{***}	-0.011^{***}
Short	0.017^{***}	0.015^{***}	0.012^{***}
Medium	-0.014^{***}	-0.016^{***}	-0.018^{***}
Long	-0.023^{***}	-0.025^{***}	-0.027^{***}

Note: 99% confidence intervals are estimated by bootstrapping predictions (1000 iterations). ***p<0.01.

Equation 3 using the same period but a year earlier (July 2017 to February 2018) and create a placebo post-BERT variable that takes on a value of 1 for November 2017 through February 2018, 0 otherwise. Since there is no algorithm change in 2017-18, this test should generate statistically insignificant results. In Tables 11 and 12 we present the results of this test. We observe that all estimates are statistically insignificant.

Table 11: Placebo Test: $log(CS)$.

	(1) 75th	(2) 50th	(3) 25th
Post × Linguistic Complexity	0.001	0.001	0.0004
	(0.001)	(0.001)	(0.001)
Post × Log(Cosine)	0.0003	0.0003	0.0004
	(0.0002)	(0.0002)	(0.0004)
Post × Linguistic Complexity × Log(Cosine)	−0.0002	−0.0002	−0.00001
	(0.0002)	(0.0003)	(0.001)
Observations	63,468	63,468	63,468
R^2	0.997	0.997	0.997

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Regressions include year-month and query fixed effects. Standard errors clustered at the query level are reported in parentheses. Data from July 2018 to February 2019.

Table 12: Placebo test: $log(CPC)$.

	(1) 75th	(2) 50th	(3) 25th
Post × Linguistic Complexity	0.001	0.002	0.001
	(0.002)	(0.002)	(0.002)
Post × Log(Cosine)	0.0004	0.0005	0.0004
	(0.001)	(0.001)	(0.001)
Post × Linguistic Complexity × Log(Cosine)	0.001	−0.0002	0.001
	(0.002)	(0.001)	(0.002)
Observations	63,468	63,468	63,468
R^2	0.979	0.979	0.979

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Regressions include year-month and query fixed effects. Standard errors clustered at the query level are reported in parentheses. Data from July 2018 to February 2019.

Changes in consumer behavior Another potential explanation for our effects is that BERT affects consumer search behavior, which, in turn, affects CPC and market size. For example, consider the case in which consumers learn that results are getting better for longer/complex queries because of BERT, and this changes the types of queries consumers are searching for. Advertises, then, may start targeting and bidding for these newly popular queries, which can lead to changes in competition and CPC.

First, it is worth noting that our analysis is over a fixed set of competitive queries. Therefore, our estimates are unlikely to capture effects involving new search queries.

Second, recall that our empirical analysis focuses on short-term effects, which reduces the likelihood that consumers will change their search behavior and advertisers will respond to these changes. Padilla et al. (2025) shows that generative LLM adoption eventually begins to shift consumer search behavior. Still, these effects take nearly 20 weeks (five months) to appear in their data, implying it takes a long time before consumers start to make meaningful adjustments.

Third, we perform a test to partially address this concern. We collect data on Google Trends search interest and search volume from SEMRush for each query in our dataset. We then use these variables to account for the popularity of each query. Then, we re-estimate Equation 3 controlling for Google Search trends interest and SEMRush's search volume. We present these results in Tables 13 and Table 14 for CS and CPC, respectively. We find that estimates remain largely consistent with our primary specification.

Table 13: Controlling for search interests and search volume: $log(CS)$

	(1)	(2)	(3)
	75th	50th	25th
Post × Linguistic Complexity	0.0042**	0.0036*	0.0036*
	(0.002)	(0.002)	(0.002)
Post × Cosine	0.0012**	0.0010*	0.0007
	(0.0005)	(0.0006)	(0.0008)
Post × Linguistic Complexity × Cosine	−0.0015**	−0.002**	−0.003***
	(0.0007)	(0.0008)	(0.0012)
Interest	0.00004	0.00004	0.00004
	(0.00004)	(0.00004)	(0.00004)
log(SV)	0.0062***	0.0062***	0.0062***
	(0.0023)	(0.0023)	(0.0023)
Observations	63,394	63,394	63,394
Note:	*p<0.1; **p<0.05; ***p<0.01		

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: All regressions include year-month and query fixed effects. Standard Errors clustered at the query level are reported in parentheses. Data is from July 2019 to February 2020.

Changes in organic rank results Another potential explanation for our results could be that advertisers respond to changes in the organic ranking results or to new features added to the results page that are a function of BERT. For example, in its release note, Google

Table 14: Controlling for search interests and search volume: $log(CPC)$

	(1) 75th	(2) 50th	(3) 25th
Post × Linguistic Complexity	-0.0254^{*}	-0.0274^{**}	-0.0290^{**}
	(0.0147)	(0.0139)	(0.0145)
Post × Cosine	0.0057^{*}	0.0066^{*}	0.0077
	(0.0032)	(0.0037)	(0.0052)
Post × Linguistic Complexity × Cosine	−0.0003	0.0006	0.00001
	(0.0051)	(0.0064)	(0.0092)
Interest	0.0001	0.0001	0.0001
	(0.0003)	(0.0003)	(0.0003)
log(SV)	0.0142	0.0142	0.0141
	(0.0129)	(0.0129)	(0.0129)
Observations	63,394	63,394	63,394
R^2	0.7951	0.7951	0.7950

Note: *p<0.1; **p<0.05; ***p<0.01

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: All regressions include year-month and query fixed effects. Standard Errors clustered at the query level are reported in parentheses. Data is from July 2019 to February 2020.

mentions that BERT will help with featured snippets. Adding more featured snippets at the top of the page may take away premium space and increase demand for sponsored search. This could lead to an increase in the number of bidders and, potentially, higher prices. Moreover, if the top organic result of a search query changes, this may suggest that BERT identified a new website as more relevant to a given query. A switch or change in the top URL could subsequently impact prices and demand for advertising space.

To address this concern, we collect monthly organic ranking data from SEMrush for the queries in our dataset. Using this data, we test for changes in the likelihood of a query resulting in a featured snippet, the number of Search Engine Results Pages (SERP) features shown on a results page, and whether the top domain has changed. Feature Snippet is an indicator variable that takes on a value of 1 if the query has a featured snippet at the top of the page, 0 otherwise. SERP Features Count is the sum of the unique number of SERP features present on the given query's results page, including featured snippets. Other SERP features include carousels, top stories, reviews, videos, and knowledge panels.[26] In general, SERP features take up top-of-page space, crowding out advertising, thus potentially affecting auctions. Finally, Domain Change is an indicator that takes the value of 1 when a month's top URL differs from the previous month's top URL, and zero otherwise.

We re-estimate Equation 3 including Featured Snippet usage, the Number of SERP features, and whether the top domain has changed compared to the previous month. We present these results in Table 15 and Table 16 . We find results that are directionally consistent but attenuated and more imprecise, particularly for CPC, where they become insignificant.[27]

Changes in advertiser spending In our model, we assume that advertisers do not adjust in the short run. This is driven by the fact that advertisers do not observe the output of BERT, and they receive limited information regarding the relevancy of their ads to queries. However, advertisers could still respond to changes in match quality or to their beliefs that match quality will improve.

We address this concern by collecting political advertiser ad spend data from the Google Ad Transparency Center (GATC). The GATC provides monthly text-spend data for political advertisers. This includes politicians, think-tanks, or any other politically affiliated organizations. We collect monthly ad spend data for advertisers who advertised between July 2019 and February 2020 and had observations both before and after October 2019. This gives us

[26]For a complete list of all features, see `https://developer.semrush.com/api/v3/analytics/basic-docs/`

[27]The loss in precision is because we do not have organic rank information for all queries (only $41,555$ out of $63,394$ observations). Indeed, we find that using the same subset of queries for which we have organic rank information and not including controls, we obtain very similar results to those reported in Table 15 and Table 16.

Table 15: Controlling for organic rank changes: $log(CS)$

	(1) 75th	(2) 50th	(3) 25th
Post × Linguistic Complexity	0.0046*	0.0039*	0.0033
	(0.0024)	(0.0023)	(0.0024)
Post × Cosine	0.0011*	0.001	0.007
	(0.0006)	(0.0007)	(0.0010)
Post × Linguistic Complexity × Cosine	−0.0011	−0.0013	−0.0016
	(0.0008)	(0.0010)	(0.0014)
log(SERP Features Count)	−0.0003	−0.0003	−0.0003
	(0.0004)	(0.0004)	(0.0004)
Feature Snippet	0.0069***	0.0069***	0.0069***
	(0.0022)	(0.0022)	(0.0022)
Domain Change	−0.0008	−0.0008	−0.0008
	(0.0008)	(0.0008)	(0.0008)
Observations	41,555	41,555	41,555
R^2	0.9772	0.9772	0.9772
Note:	*p<0.1; **p<0.05; ***p<0.01		

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: All regressions include year-month and query fixed effects. Standard errors clustered at the query level are reported in parentheses. Data is from July 2019 to February 2020.

monthly data on 256 political ad campaigns, conditional on spending on sponsored ads.

Advertising spend often has seasonal patterns. To account for this, we define two variables: $holiday_t$, which takes on a value of 1 when the month is November 2019 or December 2019, 0 otherwise, and $newyear_t$, which takes a value of 1 when the month is January 2020 or February 2020, 0 otherwise. We also introduce advertiser fixed effects to control for differences in political campaigns. We then regress ad spend on these fixed effects and plot the residuals in Figure 3. The figure shows insignificant changes in spend immediately following the introduction of BERT.

Table 16: Controlling for organic rank changes: $log(CPC)$

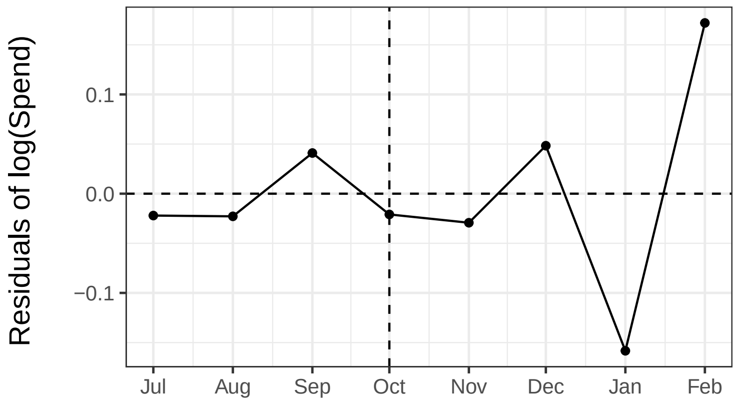
	(1) 75th	(2) 50th	(3) 25th
Post × Linguistic Complexity	−0.0158	−0.0211	−0.0206
	(0.0178)	(0.0169)	(0.0176)
Post × Cosine	0.0052	0.0050	0.0065
	(0.0039)	(0.0044)	(0.0063)
Post × Linguistic Complexity × Cosine	−0.0043	−0.0022	−0.0047
	(0.0061)	(0.0075)	(0.0104)
log(SERP Features Count)	0.0113***	0.0113***	0.0113***
	(0.0032)	(0.0032)	(0.0032)
Feature Snippet	−0.0168	−0.0167	−0.0167
	(0.0133)	(0.0133)	(0.0133)
Domain Change	−0.0078	−0.0077	−0.0077
	(0.0057)	(0.0057)	(0.0057)
Observations	41,555	41,555	41,555
R^2	0.8038	0.8038	0.8038
Note:	*p<0.1; **p<0.05; ***p<0.01		

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: All regressions include year-month and query fixed effects. Standard errors clustered at the query level are reported in parentheses. Data is from July 2019 to February 2020.

In the event study for our primary identification strategy (Table 6 and Table 5), we observe significant changes in outcomes almost immediately (November and December). In contrast, we see little visible change in Figure 3 suggesting that ad spend immediately adjusts. Combined, we interpret our results as suggestive evidence that advertiser response to BERT likely does not drive our observed empirical results.

Our descriptive finding is consistent with other empirical evidence that finds that advertisers are slow to respond to policy changes. Alcobendas and Zeithammer (2021) find that it takes months for major advertisers to change bidding strategies following a shift from a

Figure 3: Political ad spend after controlling for advertiser fixed effects, holiday (November and December), and new year (January and February)



second-price to a first-price auction format. Aridor et al. (2024) also find that advertisers take several months to adjust their advertising budgets following shifts in data privacy regulations. We certainly believe advertisers will respond and adapt over time to algorithmic changes such as BERT. However, our evidence, combined with other existing papers, suggests that this would require many more months, and it is unlikely to explain our empirical observations.

7 Conclusion

Transformers have expanded programmatic interpretation capabilities and led to the development of sophisticated LLMs. Through the "self-attention" architecture, transformers can effectively capture context in language. In online search, LLMs can significantly improve search engine interpretation capabilities. Yet, little is known about how this can impact sponsored search markets and why understanding context in language is economically meaningful.

In this paper, we consider a stylized theoretical auction model to help address this question. In our model, a seller uses an NLP algorithm to interpret queries. Both a topic and a context uniquely define queries. Context is a distinct set of linguistic information that is fundamentally different from a query's topic and is only present in specific queries. NLP algorithms generate signals across these dimensions. The search engine then uses these signals to estimate advertiser relevancy scores for sponsored search opportunities. Our model finds that when a platform improves the quality of its interpretation algorithm and starts to understand context, (1) queries generally see more bidders competing, and (2) query-level CPC can decline despite more bidders competing for ad space due to contextual understanding leading to greater relevancy score dispersion across competing bidders.

We then test our model predictions by studying how Google's October 2019 rollout of BERT affected sponsored search auction market size and CPC. We find results consistent with our model: the average number of bidders uniformly increases across queries, CPC declines for context-rich queries, and increases for context-poor queries. These findings are supported by two identification strategies and several robustness checks.

Our paper provides a rigorous theoretical and empirical study to help practitioners better understand how changes to search engine interpretation algorithms impact short-term market prices and market size. Finally, our paper contributes to the growing literature on the economic consequences of AI and LLMs in the search engine ecosystem by highlighting the surprising effects they can have on markets and the economic importance of understanding context in language.

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Online Appendix

A Web Appendix: Theory Model Proofs

Bidding Strategies Advertiser private valuations for a click are drawn $v_i \sim U[0,1]$. Conditional on being allocated to an auction, their objective is to maximize $E[v - \frac{\tilde{b}\tilde{r}}{r} | b^* r \geq \tilde{b}\tilde{r}]$, where b^* is the optimal bid, r is their relevancy score for the given auction, $\tilde{b}$ is the second highest bid value, and $\tilde{r}$ is the second highest relevancy score. ($\tilde{b}\tilde{r}$ is the second highest ad rank). They aim to maximize the following:

$$\int_0^{\frac{b^* r}{\tilde{r}}} (v - \frac{\tilde{Adr}}{r}) f(\tilde{b}) d\tilde{b}$$

Taking the derivative w.r.t b^* and rearranging, we arrive to $b^* = v$ $\square$

Platform Objective The platform wishes to maximize Equation 4.

$$\pi = I_{\{Click\}} \frac{\tilde{Adr}}{r_w} - I_{\{\theta_w \neq t\}} C \tag{4}$$

It will do so by selecting the optimal bidders given the information set $\hat{Q}_k$ it has available to it. We now go through the possible information sets the platform can receive and identify the optimal selection strategy within each set.

A.1 Full Information

With full information on both topic and context, the platform has two options: allocate only topically relevant advertisers, or allocate all advertisers with non-zero relevancy scores. Let $a = 1 - B_t$ and $b = B_t$.

A.1.1 Only Topically Relevant Allocation

With only topically relevant advertisers, there are two advertisers in the auction. One advertiser has a relevancy score of 1, the other has a relevancy score of B_t.

$$(1 - B_t)\frac{B_t}{2} + B_t[\frac{1}{2}\frac{B_t}{3} + \frac{1}{2}\frac{B_t}{3}]$$

$$\frac{B_t(1 - B_t)}{2} + B_t[\frac{B_t}{3}]$$

$$= \frac{ab}{2} + \frac{b^2}{3}$$

In this scenario, the platform allocates 2 bidders to the auction.

A.1.2 Contextually and Topically Relevant Allocation

With this scenario, the platform allocates 3 bidders to the auction. Consider the case when the $Q = (0, 0)$. The possible advertisers to allocate are $(0, 0)$, $(0, 1)$, and $(1, 0)$. (A type $(1, 1)$ advertiser has a relevancy score of 0 and will be ignored). The subsequent relevancy scores for each advertiser is 1, B_t, and $1 - B_t$. In this environment there are 5 possible cases. Define advertiser $(0, 0)$ Ad Rank as variable X, $(0, 1)$ Ad Rank as Z, and $(1, 0)$ Ad Rank as Y. $X \sim U[0, 1]$, $Z \sim U[0, b]$, and $Y \sim U[0, a]$.

With these bidders, there are five possible cases.

1. $a \leq X \leq b$, $a \leq Z \leq b$, $Y \leq a$. Occurs w.p. $\frac{(b-a)^2}{b}$

2. $X, Y, Z \leq a$. Occurs w.p. $\frac{a^2}{b}$

3. $X \geq b$. Occurs w.p. $1 - b$

4. $X, Y \leq a$, $a \leq Z \leq b$. Occurs w.p. $\frac{a(b-a)}{b}$

5. $Y, Z \leq a$, $a \leq X \leq b$. Occurs w.p. $\frac{a(b-a)}{b}$

Case 1 $a \leq X \leq b$, $a \leq Z \leq b$, $Y \leq a$. Simply a 50/50 chance either X or Z wins. If X wins, they pay Z, if Z wins, they pay X. For two uniform random variables, the expected second-highest value takes on the form $d + (m-d)\frac{n-1}{n+1}$.

$$\frac{(b-a)^2}{b}(a + (b-a)\frac{1}{3})$$
$$= \frac{a(b-a)^2}{b} + \frac{(b-a)^3}{3b}$$

Case 2 $X, Y, Z \leq a$. Each bidder has a $\frac{1}{3}$ chance of winning within this region. If the $(1,0)$ advertiser, Y, wins, the platform pays a cost of $-C$.

$$\frac{a^2}{b}(\frac{1}{3}1\frac{E[max(Z,Y)]}{1} + \frac{1}{3}b\frac{E[max(X,Y)]}{b} + \frac{1}{3}(a\frac{E[max(X,Z)]}{a} - C))$$
$$= \frac{a^2}{3b}(\frac{a}{2} + \frac{a}{2} + \frac{a}{2} - C)$$
$$= \frac{a^2}{3b}(\frac{3a}{2} - C)$$

Case 3 $X \geq b$. Occurs w.p. $1 - b = a$. In this case, X advertiser $(0,0)$ wins with certainty. The expected price to pay will be $\frac{max(Z,Y)}{1}$. In this scenario, Z and Y follow different distributions ($U[0,a]$ and $U[0,b]$). We need distribution of the max of the two variables to then calculate the expected value. Let $W = max(Z,Y)$

$$F(w) = Pr(Z \leq w)Pr(Y \leq w)$$

$$F(w) = \begin{cases} \frac{w^2}{ab} & w \leq a \\ \frac{w}{b} & a \leq w \leq b \\ 1 & w \geq b \end{cases}$$

$$f(w) = \begin{cases} \frac{2w}{ab} & w \leq a \\ \frac{1}{b} & a \leq w \leq b \\ 0 & w \geq b \end{cases}$$

$$E[W] = \int_0^b wf(w)dw = \int_0^a \frac{2w^2}{ab}dw + \int_a^b \frac{w}{b}dw$$
$$= \frac{2a^3}{3ab} + \frac{b}{2} - \frac{a^2}{2b}$$
$$= \frac{2a^2}{3b} + \frac{b}{2} - \frac{a^2}{2b}$$

This event's expected probability is a or $1 - B_t$. In this case, X wins and has a CTR and relevancy score of 1. The final expected profit in this scenario is

$$\frac{a(3b^2 - a^2)}{6b}$$

Which simplifies to

$$\frac{ab}{2} - \frac{a^3}{6b}$$

Case 4 $X, Y \leq a$, $a \leq Z \leq b$. Occurs w.p. $\frac{a(b-a)}{b}$. In this case Z advertiser $(0, 1)$ wins. Expected profits in this scenario are

$$\frac{a(b-a)}{b}E[max(X, Y)]$$

Since X and Y are both conditionally under value a, the expected max is $\frac{2a}{3}$. This gives us a final value of

$$\frac{2a^2(b-a)}{3b}$$

Case 5 This case is identical to Case 4. Final expected profits in this scenario are

$$\frac{2a^2(b-a)}{3b}$$

Expected Profit Summing across these scenarios, we reach an expected profit of

$$\frac{a^2b+b^3-a^3}{3b}-\frac{Ca^2}{3b}+\frac{ab}{2}$$

Comparison of Profits When is it better to "misallocate" only contextually relevant bidders? When the expected profits are greater compared to only allocating relevant bidders.

$$\frac{a^2b+b^3-a^3}{3b}-\frac{Ca^2}{3b}+\frac{ab}{2}\geq\frac{ab}{2}+\frac{b^2}{3}$$

Rearranging and simplifying, we find a simple solution that it is more profitable for the platform to allocate both topically and contextually relevant bidders when $B_t \geq C$.

A.2 No Information

Now, consider the case when the platform learns no information about the query. The question is whether to run an auction, and if so, how many bidders should be allocated to it? Note that because the platform has no information, all relevancy scores for all advertisers are equivalent and equal to $\frac{1}{2}$.

Consider the scenario in which the platform allocates all four advertisers and lets the market decide for itself.

$$E[\pi]=\frac{1}{4}(\frac{3}{5}+\frac{3B_t}{5}+\frac{3(1-B_t)}{5}-2C)$$

$$\frac{1}{4}(\frac{6}{5}-2C)$$

When is this greater than 0?

$$\frac{1}{4}(\frac{6}{5} - 2C) \geq 0$$

$$\frac{3}{5} \geq C$$

When $C \leq \frac{3}{5}$, the platform is willing to run an auction with all bidders. The platform has no incentive to allocate fewer advertisers, as it doesn't know who is better or worse. Randomly choosing 3 advertisers from the set of 4 leads to expected profits of $\frac{5}{24} - \frac{C}{2}$, which is always less than the profit with 4 bidders.

A.3 Partial Information

A.3.1 Topical Understanding, no Contextual Understanding

Consider now the case where the platform learns the query's topic, but not the context of the query. Here $\hat{t} = t$ and $\hat{q} = \frac{1}{2}$. Consider a query $Q = (0, 0)$ and all four advertiser $X = (0, 0)$, $Y = (0, 1)$, $Z = (1, 0)$, and $U = (1, 1)$. The relevancy scores for X and Y advertisers are $\frac{1+B_t}{2}$ and $\frac{1-B_t}{2}$ for Z and U advertisers. Topically relevant advertisers are X and Y.

Allocate Only Topically Relevant If the platform allocates only topically relevant advertisers X and Y, it creates a simple second-price auction outcome. (Relevancy scores cancel out, so the model simplifies to a simple SPA). There is a $\frac{1}{2}$ chance a given advertiser wins. If X wins, CTR is 1. If Y wins, CTR is B_t. Expected second highest value is $\frac{1}{3}$.

$$E[\pi] = \frac{1}{6} + \frac{B_t}{6}$$

$$= \frac{1 + B_t}{6}$$

Allocate All Advertisers Now, the platform could allocate all advertisers with non-zero relevancy scores. (If $B_t = 1$, Z and U advertisers are essentially removed). Define $a = \frac{1-B_t}{2}$

and $b = \frac{1+B_t}{2}$. Four possible cases could occur with all four advertisers allocated to the auction.

1. $X, Y, Z, U \leq a$. Occurs w.p. $\frac{a^2}{b^2}$
2. $X, Y \in (a, b]$ and $Z, U \leq a$. Occurs w.p. $\frac{(b-a)^2}{b^2}$
3. $X \geq a$ and $Y, Z, U \leq a$. Occurs w.p. $\frac{a(b-a)}{b^2}$
4. $Y \geq a$ and $X, Z, U \leq a$. Occurs w.p. $\frac{a(b-a)}{b^2}$

Case 1: The first case considers the scenario where all four bidder ad ranks fall under a. In this scenario, each bidder has an equal chance of winning. Define W as the expected second highest ad rank.

$$\frac{a^2}{b^2}[\frac{1}{4}X + \frac{1}{4}Y + \frac{1}{4}Z + \frac{1}{4}U]$$

$$\frac{a^2}{b^2}[\frac{1}{4}\frac{W}{b} + \frac{1}{4}\frac{W}{b}B_t + \frac{1}{4}((1-B_t)\frac{W}{a} - C) - \frac{1}{4}C]$$

$$\frac{a^2}{b^2}[W - \frac{C}{2}]$$

Now W is the expected second highest value among 4 draws from $U[0, a]$. This translates to $\frac{3a}{5}$.

$$\frac{a^2}{b^2}[\frac{3a}{5} - \frac{C}{2}]$$

Case 2: In this case, X and Y are in the range of a and b. This occurs w.p. $\frac{(b-a)^2}{b^2}$. Define W as now the second highest value among X and Y, conditional on X and Y being greater than a. Both X and Y have a relevancy score of b.

$$\frac{(b-a)^2}{b^2}[\frac{1}{2}X + \frac{1}{2}Y]$$

$$\frac{(b-a)^2}{b^2}[\frac{W}{2b} + \frac{W}{2b}B_t]$$

$$\frac{(b-a)^2}{b^2}[\frac{W}{2b}(1 + B_t)]$$

Since $2b = 1 + B_t$

$$\frac{(b-a)^2}{b^2}[W]$$

Now W is the second highest value among two draws from $U[a,b]$. This translates to $\frac{1}{3}(b-a) + a$.

$$\frac{(b-a)^2}{b^2}(\frac{b-a}{3} + a)$$

Case 3: This case occurs when $X \geq a$ and all other ad ranks are less than or equal to a. This occurs w.p. $\frac{a(b-a)}{b^2}$. In this case, X wins and pays the max among Y, Z, and U scaled by its relevancy score. Define W as the expected max among Y, Z, U. It is equal to $\frac{3a}{4}$

$$\frac{a(b-a)}{b^2}[\frac{W}{b}]$$

$$\frac{a(b-a)}{b^2}[\frac{3a}{4b}]$$

$$\frac{3a^2(b-a)}{4b^3}$$

Case 4: Case 4 is similar to case 3, except expected CTR for Y is B_t instead of 1 like it is for X. Therefore, we have

$$\frac{a(b-a)}{b^2}[\frac{3a}{4b}B_t]$$

Expected Profits This covers all the possible cases. Expected profit is now the summation of these cases. First, consider that Case 3 and 4 joined are:

$$(1 + B_t)\frac{3a^2(b-a)}{4b^3}$$

Since $2b = (1 + B_t)$, we can simplify this to

$$\frac{3a^2(b-a)}{2b^2}$$

Summing all cases now:

$$E[\pi] = \frac{a^2}{b^2}[\frac{3a}{5} - \frac{C}{2}] + \frac{(b-a)^2}{b^2}[\frac{b-a}{3} + a] + \frac{3a^2(b-a)}{2b^2}$$

What we want to know is when is this more profitable in expectation than allocating just the topically relevant advertisers.

$$\frac{a^2}{b^2}[\frac{3a}{5} - \frac{C}{2}] + \frac{(b-a)^2}{b^2}[\frac{b-a}{3} + a] + \frac{3a^2(b-a)}{2b^2} \geq \frac{b}{3}$$

Simplifying, we get:

$$b - \frac{7}{15}a \geq C$$

Since $b = \frac{1+B_t}{2}$ and $a = \frac{1-B_t}{2}$, we can further simplify

$$\frac{4 + 11B_t}{15} \geq C$$

When $C \leq \frac{4+11B_t}{15}$, it is more profitable for the company to allocate all advertisers to the auction than to run it with only topically relevant advertisers.

Allocate Topically Relevant and a Topically Irrelevant We now consider the case when the platform allocates all topically relevant advertisers and one topically irrelevant advertiser. The platform may benefit from this misallocation if it can weakly boost the final price paid by increasing the number of bidders competing. Here, they can allocate X and Y and either Z or U. If they allocate Z or U, there is a chance they are contextually relevant. Again define $a = \frac{1-B_t}{2}$ and $b = \frac{1+B_t}{2}$.

1. $X, Y \geq a$, $Z/U \leq a$. Occurs w.p. $\frac{(b-a)^2}{b^2}$
2. $X \geq a$, $Y, Z/U \leq a$. Occurs w.p. $\frac{a(b-a)}{b^2}$
3. $Y \geq a$, $X, Z/U \leq a$. Occurs w.p. $\frac{a(b-a)}{b^2}$
4. $X, Y, Z/Y \leq a$. Occurs w.p. $\frac{a^2}{b^2}$

Case 1: In this scenario, the effect of Z/U on price is irrelevant. Expected profits are the same as in Case 2 when all advertisers are allocated.

$$\frac{(b-a)^2}{b^2}(\frac{b-a}{3} + a)$$

Case 2: Here, X wins and pays the second highest price, or the max between Y and the other advertiser Z/U. Define W as the max of the two variables Y and Z/U.

$$\frac{a(b-a)}{b^2}[1\frac{W}{b}]$$

$$\frac{a(b-a)}{b^2}[\frac{2a}{3b}]$$

$$\frac{2a^2(b-a)}{3b^3}$$

Case 3: This is the same as Case 2, except expected CTR is B_t instead of 1.

$$\frac{2a^2(b-a)}{3b^3}B_t$$

Case 4: The final case considers when X, Y, and Z/U are all less than or equal to a. Define W as the expected second highest draw among the three variables between 0 and a.

$$\frac{a^2}{b^2}(\frac{1}{3}X + \frac{1}{3}Y + \frac{1}{3}Z/U)$$

$$\frac{a^2}{b^2}(\frac{1}{3}\frac{W}{b} + \frac{1}{3}\frac{W}{b}B_t + \frac{1}{3}[\frac{1}{2}(\frac{W}{a}(1 - B_t) - C) - \frac{1}{2}C])$$

$$\frac{a^2}{b^2}(\frac{W}{3b}(1 + B_t) + \frac{1}{6}(\frac{W}{a}(1 - B_t) - 2C))$$

Since $2a = (1 - B_t)$ and $2b = (1 + B_t)$, we can further simplify.

$$\frac{a^2}{b^2}(\frac{2W}{3} + \frac{1}{6}(2W - 2C))$$

$$\frac{a^2}{b^2}(W - \frac{C}{3})$$

Now W is the second highest draw of 3 draws from the distribution $U[0, a]$. This translates to $\frac{a}{2}$. This gives us:

$$\frac{a^2}{b^2}(\frac{a}{2} - \frac{C}{3})$$

Summing Up Cases

$$\frac{(b-a)^2}{b^2}(\frac{b-a}{3} + a) + \frac{2a^2(b-a)}{3b^3} + \frac{2a^2(b-a)}{3b^3}B_t + \frac{a^2}{b^2}(\frac{a}{2} - \frac{C}{3})$$

$$\frac{(b-a)^2}{b^2}(\frac{b-a}{3} + a) + \frac{4a^2(b-a)}{3b^2} + \frac{a^2}{b^2}(\frac{a}{2} - \frac{C}{3})$$

Comparison Across Selection Choices We want to know when this decision is more profitable than allocating only topically relevant or all advertisers. First, consider the case when the expected profit is greater than allocating only topically relevant advertisers.

$$\frac{(b-a)^2}{b^2}(\frac{b-a}{3}+a)+\frac{4a^2(b-a)}{3b^2}+\frac{a^2}{b^2}(\frac{a}{2}-\frac{C}{3})\geq\frac{b}{3}$$

Simplifying, we get the following:

$$\frac{1+3B_t}{4}\geq C$$

So, when $C\leq\frac{1+3B_t}{4}$, its more profitable to allocate topically relevant and one topically irrelevant advertiser to the auction compared to allocating only topically relevant.

Now, we also need to compare to allocating all advertisers.

$$\frac{(b-a)^2}{b^2}(\frac{b-a}{3}+a)+\frac{4a^2(b-a)}{3b^2}+\frac{a^2}{b^2}(\frac{a}{2}-\frac{C}{3})\geq\frac{a^2}{2b^2}[\frac{6a}{5}-C]+\frac{(b-a)^2}{b^2}[\frac{b-a}{3}+a]+\frac{3a^2(b-a)}{2b^2}$$

Simplifying this down, we get

$$\frac{3+7B_t}{10}\leq C$$

So when $\frac{3+7B_t}{10}\leq C$, its better to allocate 3 advertisers instead of all 4.

Allocating Two Topically Irrelevant and One Topically Relevant Advertiser The platform could allocate two topically irrelevant advertisers and one topically relevant advertiser. This would lead to an expected profit of:

$$E[\pi]=\frac{(b-a)}{b}\frac{2a}{3}+\frac{a}{b}[\frac{a}{2}-\frac{2C}{3}]$$

When simplifying, this translates to

$$\frac{1-B_t}{1+B_t}(\frac{2}{3}B_t + \frac{1-B_t}{4} - \frac{2}{3}C)$$

Compare it to when we allocate three bidders, where two are topically relevant and one is topically irrelevant.

$$\frac{(b-a)}{b}\frac{2a}{3} + \frac{a}{b}[\frac{a}{2} - \frac{2C}{3}] \geq \frac{(b-a)^2}{b^2}(\frac{b-a}{3} + a) + \frac{4a^2(b-a)}{3b^2} + \frac{a^2}{b^2}(\frac{a}{2} - \frac{C}{3})$$

$$0 \geq 12a^2b - 4ab^2 - 5a^3 - 3ab + 2C(2ab - a^2) + 2(b-a)^3 + 6a(b-a)^2$$

This eventually simplifies to

$$(1 - 3B_t^2 + 2B_t)C \leq \frac{B_t + B_t^2 - 5B_t^3 - 5}{4}$$

Note that between 0 and 1, $\frac{B_t+B_t^2-5B_t^3-5}{4}$ is always negative. Since $C \geq 0$ and $1 - 3B_t^2 + 2B_t \geq 0$, the left hand side is always positive, so this inequality never holds. Since allocating two topically relevant and one topically irrelevant is already trumped by alternative selections, selecting two topically irrelevant and one topically relevant is not relevant.

What Selection is Optimal Under these conditions, we now have 3 cutoff values: when $\frac{3+7B_t}{10} \leq C$ 3 bidders is better than 4, when $\frac{1+3B_t}{4} \geq C$ 3 is better than 2, and when $\frac{4+11B_t}{15} \geq C$ 4 is better than 2.

$$\frac{1+3B_t}{4} \leq \frac{4+11B_t}{15} \leq \frac{3+7B_t}{10}, \frac{1}{2} \leq B_t \leq 1$$

When $\frac{3+7B_t}{10} \leq C$, its better to allocate 2 bidders over 3 and 4 bidders. When $\frac{4+11B_t}{15} \leq C \leq \frac{3+7B_t}{10}$ its optimally to still only allocate 2 bidders. When $\frac{1+3B_t}{4} \leq C \leq \frac{4+11B_t}{15}$, it's optimal to allocate all 4 bidders. And finally, when $C \leq \frac{1+3B_t}{4}$ its optimal to also allocate all 4 bidders.

1. $\frac{3+7B_t}{10} \leq C$ Two bidders.

2. $\frac{4+11B_t}{15} \leq C \leq \frac{3+7B_t}{10}$ Two bidders.

3. $\frac{1+3B_t}{4} \leq C \leq \frac{4+11B_t}{15}$ Four bidders.

4. $C \leq \frac{1+3B_t}{4}$ Four bidders.

So the primary cutoff value is $\frac{4+11B_t}{15}$. If $C \geq \frac{4+11B_t}{15}$, the platform allocates only the topically relevant bidders, otherwise the platform allocates all bidders. $\square$

A.3.2 Contextual Understanding, no Topical Understanding

The final partial information case to consider is when the platform learns the context but not the topic of the query. Consider a query $Q = (0, 0)$ and advertisers X, Y, Z, and U where $X, Y \sim U[0, b]$ and $Z, U \sim U[0, a]$. Advertisers X and Y have relevancy scores equal to $1 - \frac{B_t}{2}$ while Z and U advertisers have relevancy scores of $\frac{B_t}{2}$.

Allocate Only Contextually Relevant Consider the case when the platform only allocates known contextually relevant advertisers (X and Y). Both advertisers have the same relevancy score. Let W equal the second highest value from the two random variables X and Y.

$$E[\pi] = [\frac{1}{2}X + \frac{1}{2}Y]$$

$$\frac{1}{2}\frac{W}{b} + \frac{1}{2}((1 - B_t)\frac{W}{b} - C)$$

$$\frac{W}{2b}(2 - B_t) - \frac{C}{2}$$

Note that $2b = 2 - B_t$

$$W - \frac{C}{2}$$

Now, $W = \frac{b}{3}$, where $b = 1 - \frac{B_t}{2}$. We therefore have:

$$E[\pi] = \frac{b}{3} - \frac{C}{2}$$

$\frac{b}{3} - \frac{C}{2}$ is the expected profit allocating only the advertisers with known context. If $C \geq \frac{2-B_t}{3}$, it is not profitable in expectation for the firm to run an auction with only contextually relevant bidders.

Allocate All Bidders An alternative option for the platform is to allocate all bidders. Let $a = \frac{B_t}{2}$ and $b = 1 - \frac{B_t}{2}$. This creates four cases.

1. $X, Y \in (a, b]$ and $Z, U \leq a$. Occurs with probability $\frac{(b-a)^2}{b^2}$
2. $X \geq a$ and $Y, Z, U \leq a$. Occurs w.p. $\frac{a(b-a)}{b^2}$
3. $Y \geq a$ and $X, Z, U \leq a$. Occurs w.p. $\frac{a(b-a)}{b^2}$
4. $X, Y, Z, U \leq a$. Occurs w.p. $\frac{a^2}{b^2}$

We now consider these cases.

Case 1: $X, Y \in (a, b]$. This occurs w.p. $\frac{(b-a)^2}{b^2}$. The expected profit is the same structure as when the platform allocates only contextually relevant bidders, except now the lower bound of values is a and not 0. Define W as the expected second highest Ad Rank among X and Y given they're greater than or equal to a.

$$\frac{(b-a)^2}{b^2}[\frac{1}{2}X + \frac{1}{2}Y]$$

$$\frac{(b-a)^2}{b^2}[\frac{1}{2}\frac{W}{b} + \frac{1}{2}((1-B_t)\frac{W}{b} - C)]$$

$$\frac{(b-a)^2}{b^2}[W - \frac{C}{2}]$$

Note that $W = \frac{1}{3}(b-a)+a$. We find that the expected profit is:

$$\frac{(b-a)^2}{b^2}[\frac{(b-a)}{3}+a-\frac{C}{2}]$$

Case 2: This occurs with $X \geq a$ and $Y \leq a$. This occurs w.p. $\frac{a(b-a)}{b^2}$. Define W as the expected max value among the three variables Y, Z, and U, given they're all less than or equal to a. Expected profit is:

$$\frac{a(b-a)}{b^2}[\frac{1}{2}\frac{W}{b}+\frac{1}{2}((1-B_t)\frac{W}{b}-C)]$$

$$\frac{a(b-a)}{b^2}[W-\frac{C}{2}]$$

Here, $W = \frac{3a}{4}$. The final expected profit in this case is

$$\frac{a(b-a)}{b^2}[\frac{3a}{4}-\frac{C}{2}]$$

Case 3: Case 3 is symmetrical to case 2, so the final expected profit in this scenario is:

$$\frac{a(b-a)}{b^2}[\frac{3a}{4}-\frac{C}{2}]$$

Case 4: The final case is when all four ad ranks are below a. This occurs w.p. $\frac{a^2}{b^2}$. Each bidder has an equal likelihood of winning.

$$\frac{a^2}{b^2}[\frac{1}{4}X+\frac{1}{4}Y+\frac{1}{4}Z+\frac{1}{4}U]$$

$$\frac{a^2}{b^2}[\frac{W}{b}+(\frac{W}{b}(1-B_t)-C)+\frac{W}{a}B_t-C]$$

Here, W is defined as the expected second highest ad rank among the four bidders, conditional on their values being less than or equal to a. This again simplifies down to $W-\frac{C}{2}$

$$\frac{a^2}{b^2}[W - \frac{C}{2}]$$

Where $W = \frac{3}{5}a$

$$\frac{a^2}{b^2}[\frac{3a}{5} - \frac{C}{2}]$$

Total Expected Profit Total profit is going to be the summation of each of these events.

$$\frac{(b-a)^2}{b^2}[\frac{(b-a)}{3} + a - \frac{C}{2}] + \frac{a(b-a)}{b^2}[\frac{3a}{4} - \frac{C}{2}] + \frac{a(b-a)}{b^2}[\frac{3a}{4} - \frac{C}{2}] + \frac{a^2}{b^2}[\frac{3a}{5} - \frac{C}{2}]$$

The question is then, when is this more profitable than allocating only contextually relevant bidders or compared to not running an auction.

$$\frac{(b-a)^2}{b^2}[\frac{(b-a)}{3} + a - \frac{C}{2}] + \frac{a(b-a)}{b^2}[\frac{3a}{4} - \frac{C}{2}] + \frac{a(b-a)}{b^2}[\frac{3a}{4} - \frac{C}{2}] + \frac{a^2}{b^2}[\frac{3a}{5} - \frac{C}{2}] \geq \frac{b}{3} - \frac{C}{2}$$

Notice that $\frac{C}{2}$ is common among all outcomes, and so the left and right sides each have a $\frac{C}{2}$, so we can cancel it out.

$$\frac{(b-a)^2}{b^2}[\frac{(b-a)}{3} + a] + \frac{a(b-a)}{b^2}[\frac{3a}{4}] + \frac{a(b-a)}{b^2}[\frac{3a}{4}] + \frac{a^2}{b^2}[\frac{3a}{5}] \geq \frac{b}{3}$$

$$\frac{(b-a)^2}{b^2}[\frac{(b-a)}{3} + a] + \frac{a(b-a)}{b^2}[\frac{3a}{2}] + \frac{a^2}{b^2}[\frac{3a}{5}] \geq \frac{b}{3}$$

$$\frac{(1+a)(b-a)^2}{3b^2} + \frac{3a^2(b-a)}{2b^2} + \frac{3a^3}{5b^2} \geq \frac{b}{3}$$

Simplifying this inequality, we find that:

$$B_t \leq \frac{15}{11}$$

In other words, it is more profitable to allocate all advertisers than to only allocate contextually relevant advertisers when $B_t \leq \frac{15}{11}$. Since $B_t \leq 1$, this always holds.

Comparing to no Auction

$$\frac{(b-a)^2}{b^2}(\frac{1+a}{3}) + \frac{a(b-a)}{b^2}\frac{3a}{2} + \frac{a^2}{b^2}\frac{3a}{5} \geq \frac{C}{2}$$

We can simplify this down to

$$C \leq \frac{10 - 30a + 45a^2 - 32a^3}{15(1-a)^2}$$

Which can be further simplified to

$$C \leq \frac{4}{(2-B_t)^2}[\frac{2}{3} - B_t + B_t^2 - \frac{4}{15}B_t^3]$$

When C is less than or equal to $\frac{4}{(2-B_t)^2}[\frac{2}{3} - B_t + B_t^2 - \frac{4}{15}B_t^3]$, it is more profitable for the company to run an auction with all bidders than to not run any auction.

Allocate Contextually Relevant and one Contextually Irrelevant Bidder Let $a = \frac{B_t}{2}$ and $b = 1 - \frac{B_t}{2}$. This creates four cases.

1. $X, Y \in (a, b]$ and $Z/U \leq a$. Occurs with probability $\frac{(b-a)^2}{b^2}$
2. $X \geq a$ and $Y, Z/U \leq a$. Occurs w.p. $\frac{a(b-a)}{b^2}$
3. $Y \geq a$ and $X, Z/U \leq a$. Occurs w.p. $\frac{a(b-a)}{b^2}$
4. $X, Y, Z/U \leq a$. Occurs w.p. $\frac{a^2}{b^2}$

Case 1:

$$E[\pi] = \frac{(b-a)^2}{b^2}(\frac{1+a}{3} - \frac{C}{2})$$

Cases 2 and 3:

$$E[\pi] = \frac{a(b-a)}{b^2}[\frac{2a}{3} - \frac{C}{2}]$$

Case 4:

$$E[\pi] = \frac{a^2}{b^2}[\frac{a}{2} - \frac{C}{2}]$$

Summing Up Cases Summing up the cases we have

$$E[\pi] = \frac{(b-a)^2}{b^2}(\frac{1+a}{3}) + \frac{a(b-a)}{b^2}(\frac{4a}{3}) + \frac{a^2}{b^2}(\frac{a}{2}) - \frac{C}{2}$$

Now we want to consider when this expected profit is greater than allocating only contextually relevant bidders or allocating all bidders. Since allocating all bidders dominates allocating only contextually relevant bidders, we will compare to allocating all bidders

Comparing to Allocating All Bidders

$$\frac{(b-a)^2}{b^2}(\frac{1+a}{3}) + \frac{a(b-a)}{b^2}(\frac{4a}{3}) + \frac{a^2}{b^2}(\frac{a}{2}) - \frac{C}{2} \geq \frac{(1+a)(b-a)^2}{3b^2} + \frac{3a^2(b-a)}{2b^2} + \frac{3a^3}{5b^2} - \frac{C}{2}$$

Here, both $\frac{C}{2}$ and $\frac{(b-a)^2}{b^2}(\frac{1+a}{3})$ cancel out. This leaves us with:

$$\frac{4a^2(b-a)}{3b^2} + \frac{a^3}{2b^2} \geq \frac{3a^2(b-a)}{2b^2} + \frac{3a^3}{5b^2}$$

$$-\frac{a^2(b-a)}{6} \geq \frac{a^3}{10}$$

$$10b \leq 4a$$

$$10(1-a) \leq 4a$$

$$\frac{5}{7} \leq a$$

$$\frac{10}{7} \leq B_t$$

Therefore, When $B_t \geq \frac{10}{7}$, the platform is better off running the auction with 3 bidders and not all 4. However, $B_t \leq 1$, so this cannot occur.

Allocate 2 Non Contextually relevant and 1 Contextually Relevant The platform could consider allocating 2 non-contextually relevant advertisers and only 1 contextually relevant advertiser to the auction. Expected profit in this case is:

$$E[\pi] = \frac{2a(b-a)}{3b} + \frac{a^2}{2b} - \frac{C}{2}$$

Comparing to allocating all four bidders:

$$\frac{2a(b-a)}{3b} + \frac{a^2}{2b} - \frac{C}{2} \geq \frac{(1+a)(b-a)^2}{3b^2} + \frac{3a^2(b-a)}{2b^2} + \frac{3a^3}{5b^2} - \frac{C}{2}$$

It is only better to allocate this combination of three bidders compared to all four when

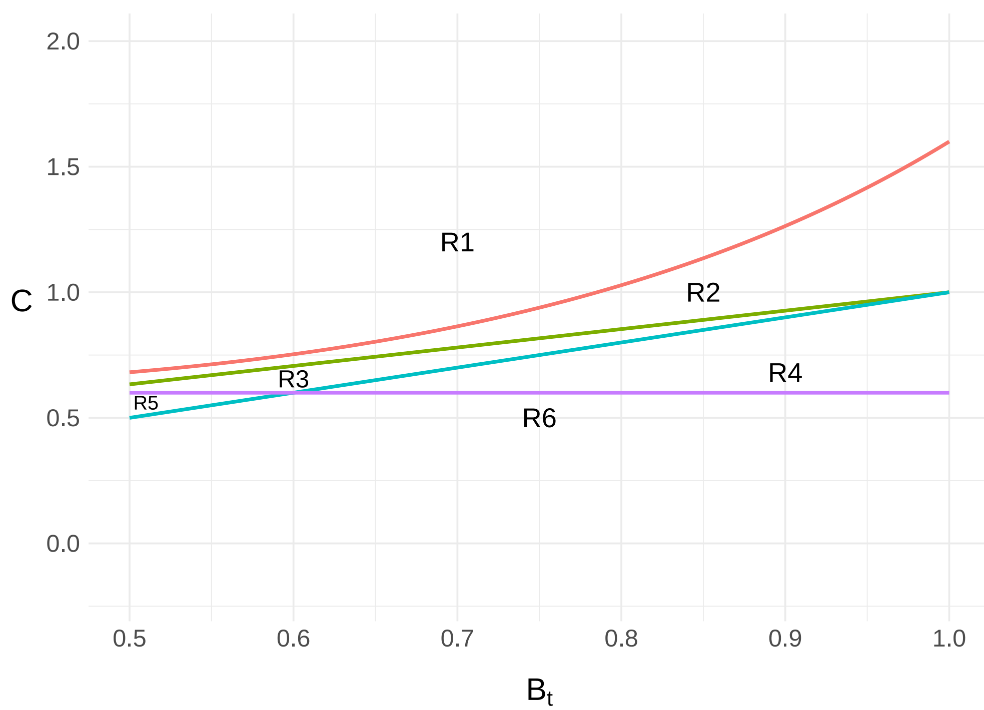
$$0 \geq 10 - 25B_t + \frac{45}{2}B_t^2 - \frac{57}{8}B_t^3$$

However, this inequality does not hold for all $\frac{1}{2} \leq B_t \leq 1$. Therefore, it is not a considered platform option.

A.4 Region Analysis

We now have our selection process criteria. Under full information, when $C \leq B_t$, it is better to allocate all topically and contextually relevant bidders. When $C \geq B_t$, the platform only wants to allocate topically relevant bidders. Under no information, the platform allocates all bidders if $C \leq \frac{3}{5}$, otherwise none are allocated. When the platform knows the topic dimension but not the context dimension, it allocates only topically relevant bidders when $C \geq \frac{4+11B_t}{15}$. Otherwise, it allocates all bidders. When context is understood, the platform allocates all bidders only when $C \leq \frac{4}{(2-B_t)^2}[\frac{2}{3} - B_t + B_t^2 - \frac{4}{15}B_t^3]$, otherwise no bidders are allocated. These inequalities create six unique regions that depend on C and B_t. In Figure 4 we plot the regions created by these thresholds.

Figure 4: Regions that determine average CPC and NOBs based on C and B_t. C is the platform's cost for showing a topically irrelevant ad, and B_t is the importance of topic alignment for a query's CTR.



Recall that w.p. γ_t, the platform learns the topic dimension of the query and w.p. γ_q, the platform learns the context dimension of the query. We now describe the expected number of bidders and CPC based on the region.

1. **Region 1:** $C \geq \frac{4}{(2-B_t)^2}[\frac{2}{3} - B_t + B_t^2 - \frac{4}{15}B_t^3]$. The platform only allocates 2 bidders when topics are known. This leads to the expected number of bidders of $2\gamma_t$.

2. **Region 2:** $C \leq \frac{4}{(2-B_t)^2}[\frac{2}{3} - B_t + B_t^2 - \frac{4}{15}B_t^3]$ and $C \geq \frac{4+11B_t}{15}$. Here, the platform

allocates only topically relevant bidders when the topic dimension is understood and all four bidders when only the context dimension is understood. This leads to an expected number of bidders of $2\gamma_t + 4\gamma_q - 4\gamma_t\gamma_q$.

3. **Region 3:** $C \leq \frac{4+11B_t}{15}$ and $C \geq B_t$. In this region, the platform allocates 2 bidders with full information, and all four bidders when the topic or context dimension is known. This leads to an expected number of bidders of $4(\gamma_t + \gamma_q) - 6\gamma_t\gamma_q$.

4. **Region 4:** $C \leq B_t$ and $C \geq \frac{3}{5}$. Here, the platform is willing to allocate all three bidders with non-zero relevancy scores with full information, and allocate all bidders when the topic or context is known. This translates to an expected number of bidders of $4(\gamma_t + \gamma_q) - 5\gamma_t\gamma_q$.

5. **Region 5:** $C \leq \frac{3}{5}$ and $C \geq B_t$. Here, the platform only wants to allocate topically relevant bidders with full information, but is willing to allocate all bidders in all other cases. This translates to an expected number of bidders of $4 - 2\gamma_t\gamma_q$.

6. **Region 6:** $C \leq \frac{3}{5}$ and $C \leq B_t$. Here, the platform allocates every bidder with non-zero relevancy scores. With full information, we assume the platform allocates all three bidders with non-zero relevancy scores. This translates to an expected number of bidders $4 - \gamma_t\gamma_q$.

Note that Google states it does not run auctions when it isn't confident it understands the query. Therefore, we can assume that the cost $C \geq \frac{3}{5}$ and the platform is unwilling to run auctions when there is no signal about the query's type.

We can observe several key points from the plot and the results. First, for queries where topic alignment is the priority (i.e., short queries where B_t is large), Regions 2 and 4 dominate most of the possible spaces. For queries where context matters more, Region 1 is taking up an increasingly larger portion of the potential space. If C is greater than 1, then all queries with B_t roughly greater than 0.7852 will be in Region 2, with all other queries in Region

1. As C lowers, we start to enter other regions. Consider when $C = 0.8$. All queries with $B_t \leq 0.647$ will live in Region 1, all queries with $0.6471 \leq B_t \leq \frac{8}{11}$ will be in Region 2, all queries where $\frac{8}{11} \leq B_t \leq 0.8$ will be in Region 3, and all queries with $B_t \geq 0.8$ will live in Region 4.

These regions define the auction environment in which a given query operates and determine how the market will respond to the introduction of a new algorithm. Note that CPC is calculated as $Pr(Click)\frac{\tilde{Adr}}{r_w}$ where $\tilde{Adr}$ is the second highest ad rank and r_w is the winning relevancy score. If no click is received, CPC is 0, if a click is received, it is the second highest ad rank scaled by the winning ad rank score. We now calculate the expected CPC and number of bidders for each region.

A.4.1 Region 1

Number of Bidders In Region 1, the platform is conservative and only allocates topically relevant bidders. Here, the expected number of bidders is $2\gamma_t$ and so the number of bidders is increasing with a since $\frac{\partial \gamma_t(a)}{\partial a} > 0$.

Cost-Per-Click (CPC) In Region 1, the platform only allocates topically relevant bidders. Expected CPC is then

$$\gamma_q\gamma_t(\frac{(1-B_t)B_t}{2} + \frac{B^2}{3}) + \gamma_t(1-\gamma_q)\frac{1+B_t}{6}$$

We can simplify to

$$\frac{\gamma_t}{6}[1 + B_t - \gamma_q(B_t - 1)^2]$$

Taking the derivative w.r.t. to a and setting less than 0, we find that CPC declines when

$$\frac{1+B_t}{6(B_t-1)^2} < \frac{\gamma_q^{'}}{\gamma_t^{'}}\gamma_t + \gamma_q$$

Notice that as B_t closes in on 1, the left-hand value approaches infinity, meaning it is unlikely that prices can decline. This aligns with our intuition that short queries, which lack context and therefore have a higher B_t value, are less likely to see prices decline because there is no context to further differentiate bidders within auctions. However, when B_t is moderate and gains to contextual understanding are large relative to topical gains, there is an opportunity for prices to decline. The intuition for the result is that when the platform's interpretation algorithm substantially improves context signals, it leads to a higher frequency of full information auction events and improved bidder differentiation. This can lead to lower prices, more relevant advertisements, and higher expected profits. (Lower prices but higher CTR).

A.4.2 Region 2

Number of Bidders In Region 2, the platform allows for some misallocation and we observe the number of bidders equal to $2\gamma_t + 4\gamma_q - 4\gamma_t\gamma_q$. Taking the derivative w.r.t. γ_q equates to $4(1-\gamma_t)$, which is always non-negative. Taking the derivative w.r.t. γ_t equates to $2 - 4\gamma_q$, which is positive when $\gamma_q < \frac{1}{2}$, 0 when $\gamma_q = \frac{1}{2}$, and negative when $\gamma_q > \frac{1}{2}$.

Taking the derivative of this w.r.t a, we find that the number of bidders will decline when:

$$(1-\gamma_t)\frac{\gamma_q^{'}}{\gamma_t^{'}} < \gamma_q - \frac{1}{2}$$

When the inequality flips, the number of bidders increases, and when there is equality, the number of bidders stays the same with a. Note that $\frac{\gamma_q^{'}}{\gamma_t^{'}}$ is always positive and that $\gamma_q - \frac{1}{2}$ only begins to be positive when $\gamma_q > \frac{1}{2}$. Therefore, when the context signal is weak ($\gamma_q < \frac{1}{2}$), it's not possible to see a decline in the number of bidders.

When $\gamma_q > \frac{1}{2}$, it is possible to see a decline in the number of bidders. Assuming the improvements to contextual signals are small (small $\gamma_q^{'}$), a decline could occur when the topic signal is also strong (large γ_t) or improvements to topic signals are large (large $\gamma_t^{'}$).

When the platform learns the topic dimension, the market consists of only two bidders. So, as this signal improves (larger γ_t), the left-hand side term shrinks due to $(1-\gamma_t)$. Whether this leads to a decline in the average number of bidders depends on the strength of the context signal. If the context signal is large and attracts all advertisers in certain situations, then we can potentially see the average number of bidders decline. Intuitively, this is capturing the market segmentation effects we might imagine could occur. The platform would, on average, remove some irrelevant advertisers and thin the market to more relevant ones.

Cost-Per-Click (CPC) Region 2 considers when the platform allocates two topically relevant bidders when the topic is known, and all four bidders when the context is known but the topic is not known. Define $J = \frac{40-60B_t+45B_t^2-16B_t^3}{30(2-B_t)^2}$. Expected CPC in Region 2 for a query is defined as

$$\frac{\gamma_t}{6}[1 + B_t - \gamma_q(B_t - 1)^2] + \gamma_q(1 - \gamma_t)A$$

To simplify notation, let $D = \frac{1+B_t}{6}$ and let $M = J + \frac{(B_t-1)^2}{6}$. The average CPC in Region 2 becomes

$$\gamma_t D + J\gamma_q - \gamma_t\gamma_q M$$

Taking the derivative w.r.t. a and finding when its less than 0, we get

$$\frac{\gamma_q^{'}}{\gamma_t^{'}}(J - M\gamma_t) < M\gamma_q - D$$

We are now interested in which cases cause this inequality to hold or not. When it does not hold, CPC can increase or stay the same.

1. $\gamma_q < \frac{D}{M}$ and $\gamma_t \leq \frac{J}{M}$. Inequality cannot hold. CPC must increase.

2. $\gamma_q < \frac{D}{M}$ and $\gamma_t > \frac{J}{M}$. Inequality can hold. CPC may decline if $\frac{\gamma_q^{'}}{\gamma_t^{'}}$ is large.

3. $\gamma_q \geq \frac{D}{M}$ and $\gamma_t < \frac{J}{M}$. Inequality can hold. CPC may decline if $\frac{\gamma_q^{'}}{\gamma_t^{'}}$ is small.

4. $\gamma_q > \frac{D}{M}$ and $\gamma_t = \frac{J}{M}$. Inequality always holds and CPC must decline. When $\gamma_q = \frac{D}{M}$ and $\gamma_t = \frac{J}{M}$, the sides equal and implies that there is no change to CPC.

5. $\gamma_q \geq \frac{D}{M}$ and $\gamma_t > \frac{J}{M}$. Inequality always holds. CPC drops with certainty.

Note that the minimum possible value of $\frac{J}{M}$ is roughly 0.971, so the topic signal for the platform must be extremely strong when the context signal is less than or equal to $\frac{D}{M}$ for prices to potentially decline. Note also that $\frac{D}{M}$ is a declining function in B_t, so higher values of B_t lead to lower thresholds $\frac{D}{M}$.

Predictions for short queries in Region 2 with introduction of BERT: $\gamma_q \geq \frac{D}{M}$ and $\gamma_t < \frac{J}{M}$. This means it's possible for CPC to decline if $\frac{\gamma_q^{'}}{\gamma_t^{'}}$. However, since we expect $\frac{\gamma_q^{'}}{\gamma_t^{'}}$ to be large, we expect that CPC will rise for short queries in Region 2.

For long queries, γ_q could either be greater than or less $\frac{D}{M}$. It is unlikely $\gamma_t > \frac{J}{M}$. More likely $\gamma_t \leq \frac{J}{M}$. In either case, it is likely that prices increase.

In Figure 5 we plot the behavior of the thresholds $\frac{D}{M}$ and $\frac{J}{M}$.

Region 2: Interesting Market Outcomes We could empirically observe a decline in CPC with more bidders if the following inequality holds:

$$\frac{\gamma_q - \frac{1}{2}}{(1-\gamma_t)} < \frac{\gamma_q^{'}}{\gamma_t^{'}} < \frac{M\gamma_q - D}{(J - M\gamma_t)}$$

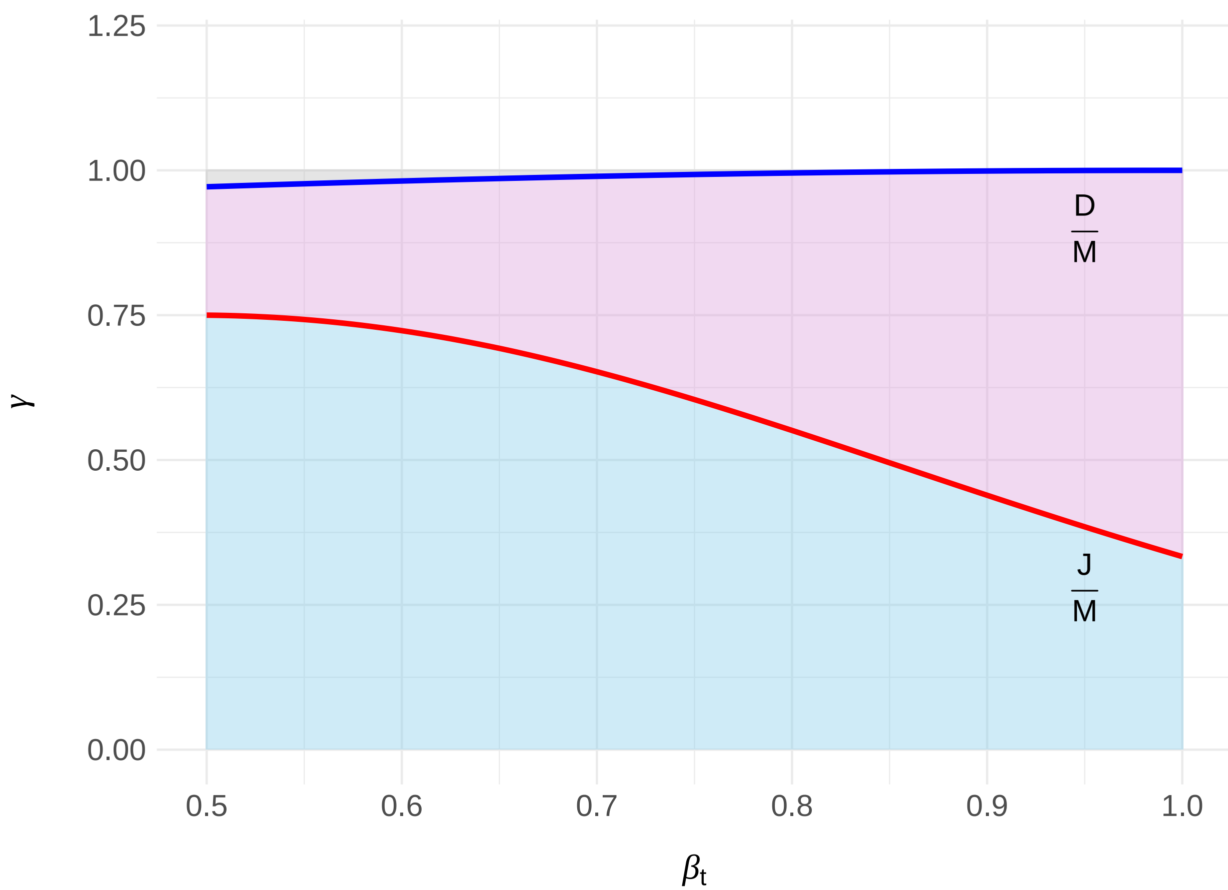
Alternatively, we could empirically observe an increase in CPC despite fewer bidders if the inequality flips:

$$\frac{\gamma_q - \frac{1}{2}}{(1-\gamma_t)} > \frac{\gamma_q^{'}}{\gamma_t^{'}} > \frac{M\gamma_q - D}{(J - M\gamma_t)}$$

A.4.3 Region 3

Number of Bidders The possibility for the number of bidders to decline when in Region 3 depends on if $\gamma_q \leq \frac{2}{3}$ and $\gamma_t > \frac{2}{3}$ or $\gamma_q > \frac{2}{3}$. This can be seen with the following inequality:

Figure 5: Region 2: Threshold functions for γ_q and γ_t that impact whether prices can increase or decrease.



$$(1 - \frac{3\gamma_t}{2})\frac{\gamma_q'}{\gamma_t'} < \frac{3}{2}\gamma_q - 1$$

The intuition for the result can be seen by looking at the expected number of bidders $4(\gamma_t + \gamma_q) - 6\gamma_t\gamma_q$. The platform typically runs auctions with all bidders when one signal is strong (topic or context), while the other is weak. However, when both signals are strong, the platform predominantly allocates resources to only the two topically relevant bidders. The number of bidders can decline when signal gains for one dimension are large, while the other dimension also has a strong signal. In a sense, the platform is moving towards a more segmented market once it has strong signals of both.

Cost-Per-Click (CPC) In Region 3, the platform allocates only topically relevant advertisers when it knows the topic and context, and all advertisers when it knows either just the topic or just the context. No advertiser is allocated when there is no information. Define $J = \frac{40-60B_t+45B_t^2-16B_t^3}{30(2-B_t)^2}$, $X = \frac{9+18B_t-3B_t^2+16B_t^3}{30(1+B_t)^2}$, and $Z = J + X - \frac{B(3-B)}{6}$. X is the expected CPC when allocating all four bidders when the platform knows the topic, and J is the expected CPC when the platform allocates all four bidders and it only knows the context dimension. Expected CPC is therefore

$$J\gamma_q + X\gamma_t - Z\gamma_t\gamma_q$$

Taking the derivative w.r.t. a and setting it less than 0, we reach the following inequality:

$$\frac{\gamma_q^{'}}{\gamma_t^{'}}(J - Z\gamma_t) < Z\gamma_q - X$$

This inequality follows a similar structure to Region 2, except Z differs from the M we defined for Region 2.

Z is going to be equal to:

$$Z = \frac{9 + 3B_t - 28B_t^2 + 11B_t^3 + 5B_t^4}{30(1 + B_t)^2} + \frac{40 - 60B_t + 45B_t^2 - 16B_t^3}{30(2 - B_t)^2}$$

which simplifies to:

$$Z = \frac{76 - 4B_t - 150B_t^2 + 173B_t^3 - 39B_t^4 - 25B_t^5 + 5B_t^6}{30(2 - B_t)^2(1 + B_t)^2}$$

The ratio $\frac{X}{Z}$ will be equal to

$$\frac{X}{Z} = \frac{(9 + 18B_t - 3B_t^2 + 16B_t^3)(2 - B_t)^2}{76 - 4B_t - 150B_t^2 + 173B_t^3 - 39B_t^4 - 25B_t^5 + 5B_t^6}$$

The ratio $\frac{J}{Z}$ will be equal to

$$\frac{J}{Z} = \frac{(40 - 60B_t + 45B_t^2 - 16B_t^3)(1 + B_t)^2}{76 - 4B_t - 150B_t^2 + 173B_t^3 - 39B_t^4 - 25B_t^5 + 5B_t^6}$$

Under what conditions will the inequality $\frac{\gamma_q^{'}}{\gamma_t^{'}}(J - Z\gamma_t) < Z\gamma_q - X$ hold

1. $\gamma_q < \frac{X}{Z}$ and $\gamma_t \leq \frac{J}{Z}$. Inequality cannot hold, and CPC must increase.

2. $\gamma_q < \frac{X}{Z}$ and $\gamma_t > \frac{J}{Z}$. Inequality can hold and prices decline if $\frac{\gamma_q^{'}}{\gamma_t^{'}}$ is large.

3. $\gamma_q > \frac{X}{Z}$ and $\gamma_t < \frac{J}{Z}$. Inequality can hold. CPC can decline if $\frac{\gamma_q^{'}}{\gamma_t^{'}}$ is small.

4. $\gamma_q > \frac{X}{Z}$ and $\gamma_t = \frac{J}{Z}$. Inequality holds, and CPC must decline. When $\gamma_q = \frac{X}{Z}$ and $\gamma_t = \frac{J}{Z}$, the sides are equal and imply there is no change to CPC.

5. $\gamma_q \geq \frac{X}{Z}$ and $\gamma_t > \frac{J}{Z}$.Inequality always holds and CPC will drop.

The behavior of CPC in Region 3 is similar to Region 2, except that the thresholds are different. In Figure 6 we present the behavior of $\frac{X}{Z}$ and $\frac{J}{Z}$.

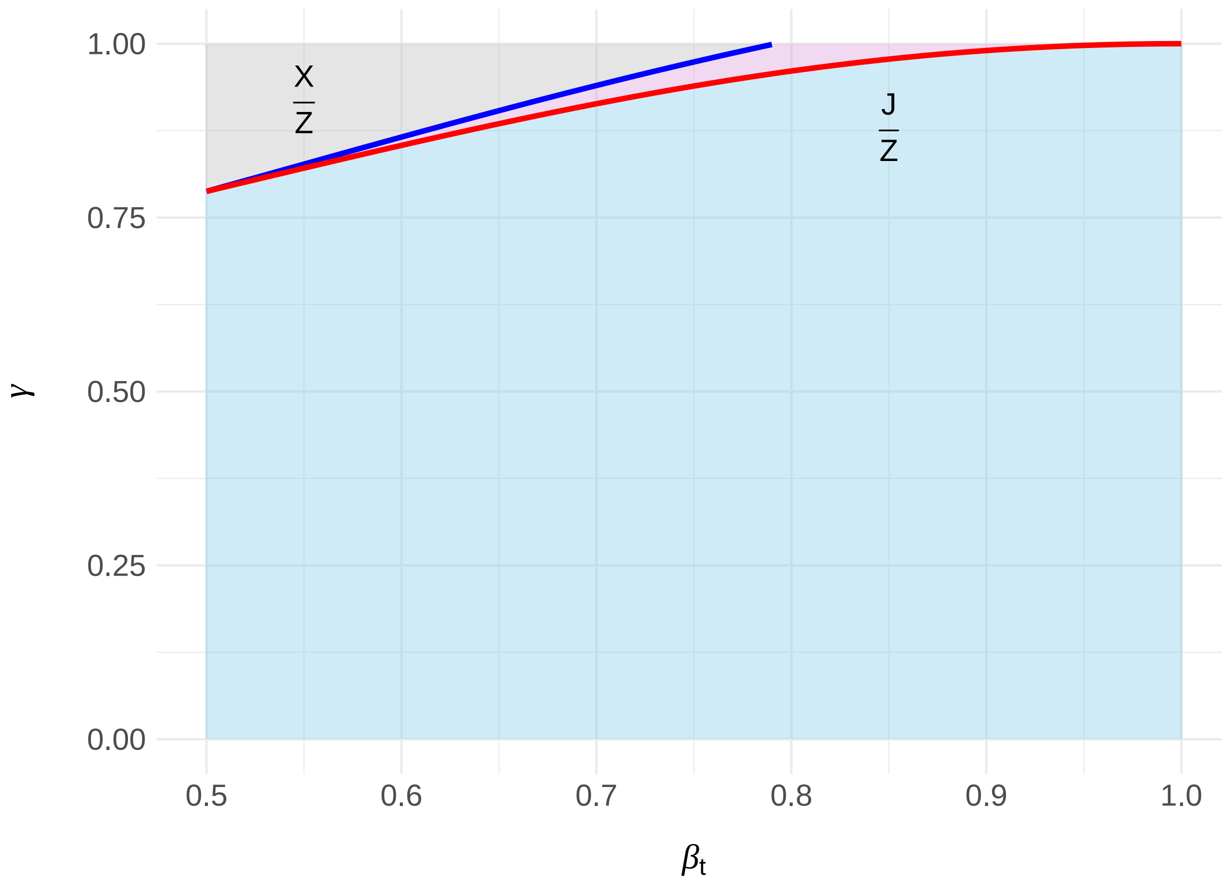
Here, the region in which γ_q and γ_t are under $\frac{X}{Z}$ and $\frac{J}{Z}$ respectively is much larger than in Region 2. The minimum value for both functions is $\frac{308}{391}$ or roughly 0.7877. If γ_q and γ_t are both less than that value, then prices cannot decline in Region 3. If we believe $\gamma_t > \frac{J}{Z}$ and $\gamma_q < \frac{X}{Z}$ and $\frac{\gamma_q^{'}}{\gamma_t^{'}}$ is large then prices are likely to decline. This may occur for longer queries with smaller B_t.

Region 3: Interesting Market Outcomes We could empirically observe a decline in CPC with more bidders if the following inequality holds:

$$\frac{\frac{3}{2}\gamma_q - 1}{1 - \frac{3\gamma_t}{2}} < \frac{\gamma_q^{'}}{\gamma_t^{'}} < \frac{Z\gamma_q - X}{(J - Z\gamma_t)}$$

Alternatively, we could empirically observe an increase in CPC despite fewer bidders if the inequality flips:

Figure 6: Region 3: Threshold functions for γ_q and γ_t that impact whether prices can increase or decrease.



$$\frac{\frac{3}{2}\gamma_q - 1}{1 - \frac{3\gamma_t}{2}} > \frac{\gamma_q^{'}}{\gamma_t^{'}} > \frac{Z\gamma_q - X}{(J - Z\gamma_t)}$$

A.4.4 Region 4

Number of Bidders Region 4 behaves similarly to Region 3, except that the threshold values are higher. Now, $\gamma_q \leq \frac{4}{5}$ and $\gamma_t > \frac{4}{5}$ or $\gamma_q > \frac{4}{5}$ can lead to a decline in the number of bidders. Otherwise, the number of bidders increases when the inequality is flipped and remains zero at equality.

$$(1 - \frac{5\gamma_t}{4})\frac{\gamma_q^{'}}{\gamma_t^{'}} < \frac{5}{4}\gamma_q - 1$$

In this region, the likelihood of seeing a decline in the number of bidders is smaller because the number of bidders is generally higher due to three bidders being allocated with full information instead of only two.

Cost-Per-Click (CPC) Region 4 differs from Region 3 because the platform is willing to allocate the three advertisers with non-zero relevancy scores to the auction, not just the topically relevant advertisers. Like Region 3, define $J = \frac{40-60B_t+45B_t^2-16B_t^3}{30(2-B_t)^2}$ and $X = \frac{9+18B_t-3B_t^2+16B_t^3}{30(1+B_t)^2}$. X is the expected CPC when allocating all four bidders when the platform knows the topic and J is the expected CPC when the platform allocates all four bidders and it only knows the context dimension. Finally, define $P = \frac{-5+20B_t-25b_t^2+15B_t^3-3b_t^4}{6B_t}$ and $L = X + J - P$. P is the expected CPC when three bidders are allocated to the auction when the platform has full information. CPC falls when:

$$\frac{\gamma_q^{'}}{\gamma_t^{'}}(J - L\gamma_t) < L\gamma_q - X$$

This is the same structure as Region 3, except L replaces Z.

Under what conditions will the inequality $\frac{\gamma_q^{'}}{\gamma_t^{'}}(J - L\gamma_t) < L\gamma_q - X$ hold

1. $\gamma_q < \frac{X}{L}$ and $\gamma_t \leq \frac{J}{L}$. Inequality cannot hold, and CPC must increase.

2. $\gamma_q < \frac{X}{L}$ and $\gamma_t > \frac{J}{L}$. Inequality can hold and prices decline if $\frac{\gamma_q^{'}}{\gamma_t^{'}}$ is large.

3. $\gamma_q > \frac{X}{L}$ and $\gamma_t < \frac{J}{L}$. Inequality can hold. CPC can decline if $\frac{\gamma_q^{'}}{\gamma_t^{'}}$ is small.

4. $\gamma_q > \frac{X}{L}$ and $\gamma_t = \frac{J}{L}$. Inequality holds, and CPC must decline. When $\gamma_q = \frac{X}{L}$ and $\gamma_t = \frac{J}{L}$, the sides are equal and imply there is no change to CPC.

5. $\gamma_q \geq \frac{X}{L}$ and $\gamma_t > \frac{J}{L}$.Inequality always holds and CPC will drop.

L is equal to:

$$L = \frac{100 - 224B_t + 81B_t^2 + 340B_t^3 - 282B_t^4 - 119B_t^5 + 230B_t^6 - 105B_t^7 + 15B_t^8}{30(2 - B_t)^2 B_t (1 + B_t)^2}$$

The ratio $\frac{J}{L}$ is equal to

$$\frac{J}{L} = \frac{(B_t(1 + B_t)^2)(40 - 60B_t + 45B_t^2 - 16B_t^3)}{100 - 224B_t + 81B_t^2 + 340B_t^3 - 282B_t^4 - 119B_t^5 + 230B_t^6 - 105B_t^7 + 15B_t^8}$$

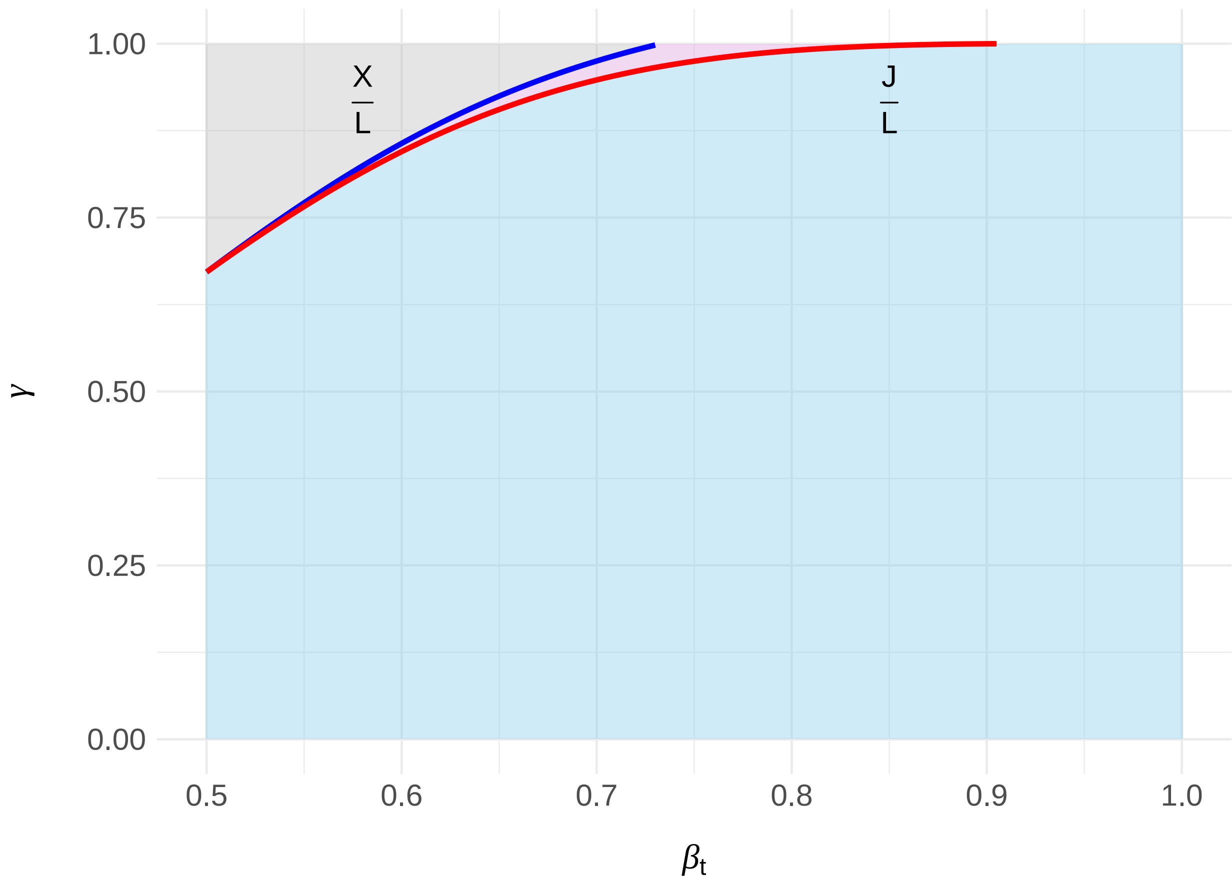
and the ratio $\frac{X}{L}$ is equal to

$$\frac{X}{L} = \frac{(9 + 18B_t - 3B_t^2 + 16B_t^3)(B_t(2 - B_t)^2)}{100 - 224B_t + 81B_t^2 + 340B_t^3 - 282B_t^4 - 119B_t^5 + 230B_t^6 - 105B_t^7 + 15B_t^8}$$

In Figure 7 we plot $\frac{X}{L}$ and $\frac{J}{L}$. Note that the gray area is slightly larger compared to Region 3.

Region 4: Interesting Market Outcomes We could empirically observe a decline in CPC with more bidders if the following inequality holds:

Figure 7: Region 4: Threshold functions for γ_q and γ_t that impact whether prices can increase or decrease.



$$\frac{\frac{5}{4}\gamma_q - 1}{1 - \frac{5\gamma_t}{4}} < \frac{\gamma_q^{'}}{\gamma_t^{'}} < \frac{L\gamma_q - X}{(J - L\gamma_t)}$$

Alternatively, we could empirically observe an increase in CPC despite fewer bidders if the inequality flips:

$$\frac{\frac{5}{4}\gamma_q - 1}{1 - \frac{5\gamma_t}{4}} > \frac{\gamma_q^{'}}{\gamma_t^{'}} > \frac{L\gamma_q - X}{(J - L\gamma_t)}$$

A.4.5 Regions 5 and 6

Number of Bidders These regions experience a decline in the number of bidders due to the platform allocating fewer bidders when it has full information. The number of bidders declines with a. The decline is faster in Region 5 compared to Region 6.

Cost-Per-Click (CPC): Region 5 Under Region 5, it is optimal for the platform to allocate all bidders except when it has full information. With full information, it allocates only the topically relevant bidders (2). Region 5 behaves similarly to Region 3. Define $F = \frac{3}{10}$. This is the average CPC when the platform allocates all bidders when it has no information. Again, define $J = \frac{40-60B_t+45B_t^2-16B_t^3}{30(2-B_t)^2}$, $X = \frac{9+18B_t-3B_t^2+16B_t^3}{30(1+B_t)^2}$, and $Z = J + X - \frac{B(3-B)}{6}$. Prices can decline if the following inequality holds:

$$\frac{\gamma_q^{'}}{\gamma_t^{'}}(J - Z\gamma_t - F(1+\gamma_t)) < Z\gamma_q - X + F(1-\gamma_q)$$

This is similar to the inequality in Region 3, except we have this extra $-F(1+\gamma_t)$ on the left-hand side and $F(1-\gamma_q)$ on the right-hand side. Ultimately, we find that critical inequality thresholds are now $\frac{J-F}{Z-F}$ for γ_t and $\frac{X-F}{Z-F}$ for γ_q. These are the same critical thresholds in Region 3, except both the numerator and denominator are reduced by F.

Note that the max value of J is $\frac{3}{10}$ when $B_t = 1$. Therefore $\frac{J-F}{Z-F} \leq 0$, meaning $\gamma_t \geq \frac{J-F}{Z-F}$ will always hold. When $B_t \leq 0.75$, $\gamma_q \geq \frac{X-F}{Z-F}$ because $\frac{X-F}{Z-F} \leq 0$. Once $B_t > 0.75$, γ_q can potentially be less than $\frac{X-F}{Z-F}$, though it must be extremely small.

In general, CPC is going to decline in Region 5 with the decline in the number of bidders. The market segmentation effects drive this due to context.

Cost-Per-Click (CPC): Region 6 The relevant threshold inequality for Region 6 is similar to Region 4 and Region 5. Again, define $J = \frac{40-60B_t+45B_t^2-16B_t^3}{30(2-B_t)^2}$, $X = \frac{9+18B_t-3B_t^2+16B_t^3}{30(1+B_t)^2}$, $P = \frac{-5+20B_t-25b_t^2+15B_t^3-3b_t^4}{6B_t}$, and $L = X + J - P$.

$$\frac{\gamma_t^{'}}{\gamma_q^{'}}(J - L\gamma_t - F(1+\gamma_t)) < L\gamma_q - X + F(1-\gamma_q)$$

Now, the relevant thresholds are $\frac{J-F}{L-F}$ for γ_t and $\frac{X-F}{L-F}$ for γ_q. Whether prices decline or increase, they follow the same constraints as Region 4, except the thresholds are now $\frac{J-F}{L-F}$ and $\frac{X-F}{L-F}$. Like Region 5, $\gamma_t \geq \frac{J-F}{Z-F}$ will always hold and when $B_t \leq 0.75$, $\gamma_q \geq \frac{X-F}{Z-F}$ because $\frac{X-F}{Z-F} \leq 0$. Once $B_t > 0.75$, γ_q can potentially be less than $\frac{X-F}{Z-F}$, though it must be extremely small.

A.5 Average CPC and Number of Bidders Summary

Below are the full set of results for the expected number of bidders and CPC by region given C and B_t. Let $J = \frac{40-60B_t+45B_t^2-16B_t^3}{30(2-B_t)^2}$, $X = \frac{9+18B_t-3B_t^2+16B_t^3}{30(1+B_t)^2}$, $Z = J + X - \frac{B(3-B)}{6}$, $P = \frac{-5+20B_t-25B_t^2+15B_t^3-3B_t^4}{6B_t}$, and $L = X + J - P$.

1. **Region 1:** $C \geq \frac{4}{(2-B_t)^2}[\frac{2}{3} - B_t + B_t^2 - \frac{4}{15}B_t^3]$

    - $E[NOB] = 2\gamma_t$
    - $E[CPC] = \frac{\gamma_t}{6}[1 + B_t - \gamma_q(B_t-1)^2]$

2. **Region 2:** $\frac{4+11B_t}{15} \leq C \leq \frac{4}{(2-B_t)^2}[\frac{2}{3} - B_t + B_t^2 - \frac{4}{15}B_t^3]$

    - $E[NOB] = 2\gamma_t + 4\gamma_q - 4\gamma_t\gamma_q$
    - $E[CPC] = \gamma_t\frac{1+B_t}{6} + J\gamma_q - \gamma_t\gamma_q(J + \frac{(B_t-1)^2}{6})$

3. **Region 3:** $B_t \leq C \leq \frac{4+11B_t}{15}$

- $E[NOB] = 4(\gamma_t + \gamma_q) - 6\gamma_t\gamma_q$
- $E[CPC] = J\gamma_q + X\gamma_t - Z\gamma_t\gamma_q$

4. **Region 4:** $\frac{3}{5} \leq C \leq B_t$
    - $E[NOB] = 4(\gamma_t + \gamma_q) - 5\gamma_t\gamma_q$
    - $E[CPC] = X\gamma_t + J\gamma_q - L\gamma_t\gamma_q$
5. **Region 5:** $B_t \leq C \leq \frac{3}{5}$
    - $E[NOB] = 4 - 2\gamma_t\gamma_q$
    - $E[CPC] = (J - \frac{3}{10})\gamma_q + (X - \frac{3}{10})\gamma_t - (Z - \frac{3}{10})\gamma_t\gamma_q$
6. **Region 6:** $C \leq \frac{3}{5}$ and $C \leq B_t$
    - $E[NOB] = 4 - \gamma_t\gamma_q$
    - $E[CPC] = (X - \frac{3}{10})\gamma_t + (J - \frac{3}{10})\gamma_q - (L - \frac{3}{10})\gamma_t\gamma_q$

B Web Appendix: Sampling Procedure

Below is the list of search topics we asked for in our Amazon MTurk survey.

Insurance, Travel, Home Renovation, Cars, Restaurants, How-to's, Things To Do, Company Research, Product Research, Financial Resources, Financial Products, News, Politics, Animals, Books, Movies, Shows, Clothes, Electronics, Pest Control, Birthday Gifts, Kitchen Shopping, Moving Services, Social Media, Entertainment, Cooking, Transportation, Home Decor, Health, Medicine, Cosmetics.

Each participant was randomly asked about only five of these categories.

C Primary Identification Strategy: Additional Robustness Checks

C.1 Alternative Measures

Continuous Linguistic Complexity results Our primary LC measure is binary, allowing for straightforward interpretation and presentation of results. We also model using a continuous form of LC. We create a binary indicator for whether a query's syntax tree is greater than the median and a binary indicator for whether it's POS count is greater than the median. We then take the sum of these two observations, meaning a query's continuous LC measure can have a value of 0, 1, or 2. 0 means both the POS count and syntax tree depth were below the dataset median, while 2 means both were above the median. Using this continuous form of LC, we present results for CS in Table 17 and results for CPC in Table 18.

Table 17: $log(CS)$ changes with Cosine and Continuous Linguistic Complexity.

	(1)	(2)	(3)
	75th	50th	25th
Post × Continuous LC	0.009^{***}	0.007^{***}	0.007^{***}
	(0.002)	(0.002)	(0.002)
Post × Log(Cosine)	0.001^{***}	0.001^{**}	0.001
	(0.0005)	(0.001)	(0.001)
Post × Continuous LC × Log(Cosine)	-0.002^{***}	-0.003^{***}	-0.004^{***}
	(0.001)	(0.001)	(0.001)
Observations	63,466	63,466	63,466
R^2	0.980	0.979	0.979

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: All regressions include year-month and query fixed effects. Standard errors clustered at the query level are reported in parentheses. Data is from July 2019 to February 2020.

Table 18: $log(CPC)$ changes with Cosine and Continuous Linguistic Complexity.

	(1)	(2)	(3)
	75th	50th	25th
Post × Continuous LC	-0.037^{**}	-0.040^{***}	-0.042^{***}
	(0.016)	(0.015)	(0.016)
Post × Log(Cosine)	0.005	0.006^{*}	0.006
	(0.003)	(0.004)	(0.005)
Post × Continuous LC × Log(Cosine)	0.001	0.003	0.003
	(0.006)	(0.008)	(0.011)
Observations	63,466	63,466	63,466
R^2	0.795	0.795	0.795

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: All regressions include year-month and query fixed effects. Standard errors clustered at the query level are reported in parentheses. Data is from July 2019 to February 2020.

Alternative Cosine measure We can standardize the cosine similarity scores within each vector space and compare differences across the vector spaces within a given query. Therefore, for a given query, we can estimate its position in the BERT cosine similarity distribution and the D2V cosine similarity distribution, and calculate the difference (or change) when moving from D2V to BERT. This allows us to analyze queries that, when transitioning from the D2V vector space to the BERT vector space, show a comparative increase (or decrease) in similarity with all other queries in our dataset. This measure can identify which queries are potentially more or less related to other queries and topics. We are theoretically interested in this measurement because we hypothesize that increased relatedness leads to an increase in the number of advertisers deemed "related", leading to an increase in bidders (CS). The downside of measuring these changes is that it doesn't capture shifts in the related queries like our first measurement. The average cosine similarity difference between the two normalized counts is 0. At 0, there was no relative shift in the cosine similarity. Our primary focus is to understand what happens when cosine-relatedness increases. To operationalize this, we create a binary variable called Normalized Cosine Diff, which takes on the value of 1 if the query's relative change is greater than the 75th percentile (0.8532), and zero otherwise. Doing so, we can compare queries with a significant increase in cosine similarity to those with comparatively little change or a reduction in cosine similarity densities.

Table 19: $log(CS)$ and $log(CPC)$ changes with Cosine Quartile and Linguistic Complexity.

	(1)	(2)
	Log(CS)	Log(CPC)
Post × Linguistic Complexity	0.003^{*}	-0.029^{**}
	(0.002)	(0.013)
Post × Top Quartile Cosine Score	0.005^{*}	0.031^{*}
	(0.003)	(0.016)
Post × Linguistic Complexity × Top Quartile Cosine Score	-0.008^{**}	0.004
	(0.003)	(0.024)
Observations	63,466	63,466
R^2	0.979	0.795

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: All regressions include year-month and query fixed effects. Standard errors clustered at the query level are reported in parentheses. Data is from July 2019 to February 2020.

We also estimate Equation 3 using the previously described quartile indicator variable as an alternative cosine measure. Recall that the Top Quartile Cosine Score takes on a value of 1 if the relative change in the difference between the normalized BERT and normalized D2V cosine ranks is greater than the 75th percentile of differences, zero otherwise. In other words, the Top Quartile Cosine Score takes on a value of 1 when a query sees a large increase in the relative similarity to it and other queries when shifting from the D2V vector space to the BERT vector space. This provides us with a different way to measure increases or decreases in "relatedness" to other queries. We find directionally consistent results with those found in the models shown in Tables 7 and 9. Under this model specification, the average predicted values at the aggregate level indicate that CS increases by approximately 0.5% and CPC decreases by 0.9%. Short queries see CS increase by roughly 0.2% and CPC increase by 1.5%, medium queries see CS increase by 0.6% and CPC decrease by 1.7%. Finally, long queries see CS increase by 0.6% while CPC decreases by 2.7%. Again, all estimates are directionally consistent with our primary DiD specification but are more conservative.

Table 20: $log(CS)$ and $log(CPC)$ changes with Cosine Quartile and Continuous Linguistic Complexity.

	(1)	(2)
	Log(CS)	Log(CPC)
Post × Continuous LC	0.006***	−0.040***
	(0.002)	(0.014)
Post × Top Quartile Log(Cosine) Score	0.005**	0.025
	(0.002)	(0.015)
Post × Continuous LC × Top Quartile Log(Cosine) Score	−0.011***	0.020
	(0.004)	(0.030)
Observations	63,466	63,466
R^2	0.979	0.795

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: All regressions include year-month and query fixed effects. Standard errors clustered at the query level are reported in parentheses. Data is from July 2019 to February 2020.

D Web Appendix: Alternative Identification Strategy

Our secondary identification strategy exploits the panel nature of our data and compares changes in each query's CS and CPC before and after the introduction of BERT with changes in the same query's CS and CPC in the year before the introduction of BERT.[28] In doing so, we implement a strategy akin to a difference-in-differences (DD) where the treated and control queries are the same, but outcomes are observed across years.[29] Our identification strategy is similar to those employed in recent papers, such as Eichenbaum et al. (2020), Bollinger et al. (2022), and Liaukonyte et al. (2022).

This strategy aims to use variation across time within queries to identify BERT. An advantage of this strategy is that selection into the treatment is not an issue since treated and control queries are the same. However, time-varying unobservables that vary at the year-month level may bias our results. For example, our results could be positively biased if some market factor correlated with CS and/or CPC and motivated Google to launch BERT in October 2019. Alternatively, queries may not be comparable across years if a query correlates with year-month-specific unobservable events, e.g., large concerts or sporting events. Both issues are classic DD challenges akin to unit-time-specific events tampering with causal effect estimates.

We operationalize this identification strategy using the following model:

$$Y_{i,j,t} = \beta_1 Treated_{i,j} \times Post_t + \delta_i + \gamma_j + \psi_t + \epsilon_{i,j,t}, \tag{5}$$

where $Y_{i,j,t}$ is the outcome of interests (either $log(CPC)ijt$ or $log(CS)_{ijt}$) for query i, year group j and month t.[30] $Treated_{ij}$ is a binary indicator that takes on the value of one when query i is observed during the treated year-group window (July 2019–February 2020), zero

[28] Essentially, we are comparing outcomes for each query in the year in which BERT was introduced with outcomes in the year prior to BERT introduction.

[29] In standard DD settings, outcomes are observed over the same period, but treated and control units are different.

[30] We add 1 to both CPC and CS to avoid taking the log of zero.

otherwise (July 2018–February 2019).[31] $Post_t$ is a dummy that takes on the value one if the month t is after the month Google introduced BERT (November–February), zero otherwise (July–October). We include query fixed effects (δ_i) to control for time-invariant query unobservables, year-group fixed effects (γ_j) to control for group-specific shocks impacting all queries in the same year-group (e.g., in 2019–20, demand is higher for search ads), and month-of-year fixed effects (ψ_t) to account for monthly shocks common to all queries (e.g., holidays and advertiser monthly spend). We cluster standard errors at the query level to account for potential serial correlations in the dependent variable and estimate the model using OLS. The parameter of interest is β_1, representing the incremental impact of BERT on the sponsored search auction outcomes of interest.

We estimate Equation 5 from July to February so that we have four pre-BERT months (July–October) and four post-BERT months (November–February). In addition, we focus on two periods, July 2018–February 2019 (control period) and July 2019–February 2020 (treated period). In Section D.2, we show that results hold with additional control year groups, specifically, July 2016–February 2017 and July 2017–February 2018 months.

Identification Checks The key assumption behind our DD identification strategy is that no unobserved time-variant, group-specific shocks correlate with the entry of BERT and auction market outcomes.

In our setting, we worry about changes in advertiser or consumer search behavior due to time-varying unobservable events that correlate with BERT's release. It is possible that consumer search demand characteristics differ across years and that consumer behavior or advertiser behavior in, say, the holiday season in 2018 is different than that in 2019. If this is the case, the results we observe may be due to these unobservable time-varying factors. This concern is analogous to worrying about unit-specific time-varying unobservable events correlated with the treatment event in a DD across states or cities.

[31] We refer to j as a group rather than a year because our treated (control) periods fall across multiple years. For example, the treated period includes the years 2019 and 2020 because the treated period spans July 2019 to February 2020.

To reduce concerns about these issues, we do two things. First, to alleviate demand-side consumer shocks, our sampling procedure removes queries that have intermittent or volatile search volumes that may correlate with unobservable time-varying events. For example, a query about a political event (e.g., November 2018 elections) that may see a spike in search in a particular month and then quickly disappear will not be part of our sample. Doing this restricts our analysis to persistently searched queries in order to minimize empirical exposure to volatile, time-varying search queries. Second, as is common in DD analyses, we show that treated and control units behaved similarly before the introduction of BERT (parallel trends assumption), suggesting that the year prior to BERT is a good counterfactual. We do so by implementing an event study design.

To obtain event study parameter estimates, we estimate the following specifications:

$$Y_{i,j,t} = \beta_1 Treated_{i,j} \times Month_t + \delta_i + \gamma_j + \psi_t + \epsilon_{i,j,t}, \tag{6}$$

Everything is as in Equation 5, but we replace our binary $Post_t$ variable with eight monthly dummies that indicate the eight-month window from July to February. We estimate Equation 6 for all the outcomes we study, setting the baseline level for the monthly dummies to be October (the month prior to the introduction of BERT). In the next section, we present these event studies along with the main estimates.

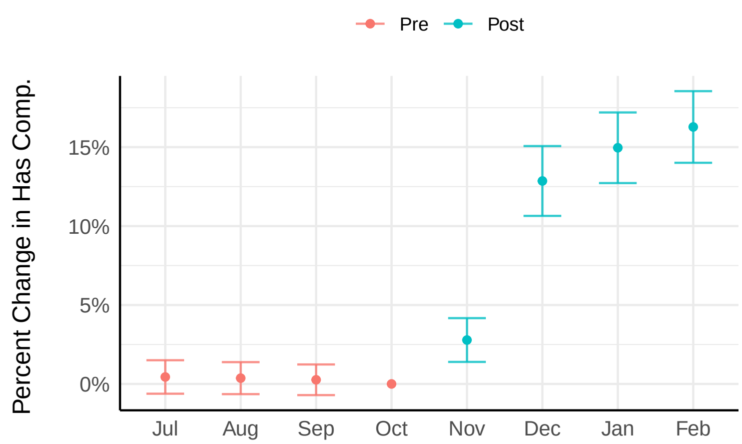
D.1 Results

Market size results The theoretical model predicts that topic signal gains due to BERT will help Google expand auction markets. This will lead to Google identifying relevant bidders more often for query auctions.

To test this prediction, we start by splitting the data into competitive and non-competitive queries. We deem a query non-competitive if the mode Competition Score during the July-October pre-period in 2018 and 2019 is zero. We find that roughly 20% of our queries are deemed non-competitive.

We then take these queries deemed non-competitive and create a binary variable called $\textit{Has Competition}_{ijt}$ that takes on the value one if query i's Competition Score is greater than zero for a given year-group j month t, zero otherwise. Using this variable, we can measure what proportion of non-competitive auctions become competitive in the post-BERT periods. We then estimate Equations 5 and 6, using $\textit{Has Competition}_{ijt}$ as the dependent variable. In Figure 8, we visualize the event study parameter estimates. We find that prior to the treatment, estimates are indistinguishable from zero, with no apparent trends. In the post-treatment period, we observe that the estimates become positive, an effect consistent with BERT increasing ad space supply by identifying more relevant bidders for auction opportunities.

Figure 8: The effect of BERT on Has Competition for non-competitive queries. Error bars represent 95% confidence intervals.



We present the estimates of Equation 5 in Table 21. We find that BERT converts roughly 11.5% of non-competitive queries into competitive markets.[32] This suggests that topical signal improvements due to BERT help the platform identify meaningfully relevant

[32] As a sanity check, we estimated Equation 5 using $log(CPC)$ as the dependent variable for the queries deemed non-competitive before BERT. We find a positive and statistically significant coefficient (0.03 with $p = 0.0005$), suggesting that CPC also rises with the increase in the number of competitive auctions for the non-competitive queries following the introduction of BERT.

advertisers for more query auctions.

Table 21: The effect of BERT on Has Competition for non-competitive queries.

	(1)
Post × Treated	0.114^{***}
	(0.009)
Observations	30,474
R^2	0.380

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Regressions include year-group, month, and query fixed effects. Standard errors clustered at the query level are reported in parentheses.

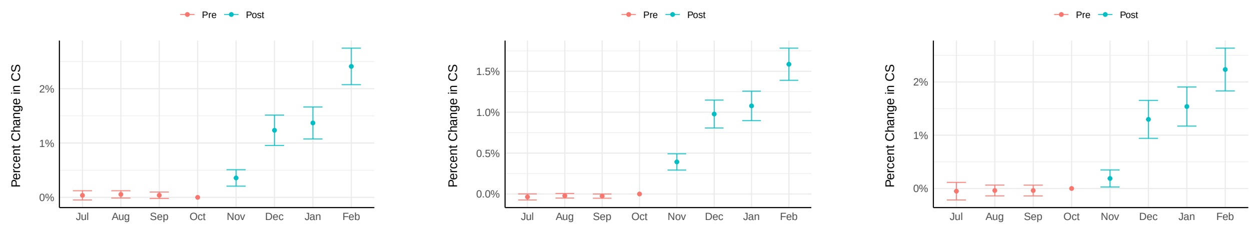
Our theoretical model predicts that under sufficiently large C, the average number of bidders should generally increase across B_t. To test variation in B_t, we group queries based on their length. Query length correlates with the degree of contextual information present within the query (high B_t associated with short queries, longer queries associated with a smaller B_t) and is often used as a rule-of-thumb measure by advertisers when defining targeting strategies. We then estimate Equation 5 and Equation 6 by query length and using $log(CS)$ as the dependent variable.

We start by presenting the event study estimates in Figure 9. Before the treatment, we find estimates close to zero, partially validating the parallel trends assumption. In the post-treatment period, estimates become positive and significant for short, medium, and long queries, suggesting that competition increases for all queries that were competitive prior to BERT's introduction.

In Table 22, we report the main effect estimates from Equation 5. We find CS increases between 1% and 1.3% for all queries.

Consistent with our model's predictions, the empirical evidence suggests that BERT's introduction helped Google uniformly expand auction markets and improve advertiser allo-

Figure 9: Percent change in Competition Score by query length. Error bars represent 95% confidence intervals.



(a) Short queries (b) Medium queries (c) Long queries

Table 22: The effect of BERT on $log(CS)$ by query length.

	(1) Short	(2) Medium	(3) Long
Post × Treated	0.013***	0.010***	0.013***
	(0.001)	(0.001)	(0.001)
Observations	47,912	88,888	23,887
R^2	0.951	0.971	0.920

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Estimated with competitive queries. Regressions include year-group, month, and query fixed effects. Standard errors clustered at the query level are reported in parentheses.

cation to auctions. This is because BERT helped Google overcome the negative cost C and do a better job identifying topically relevant advertisers for queries.

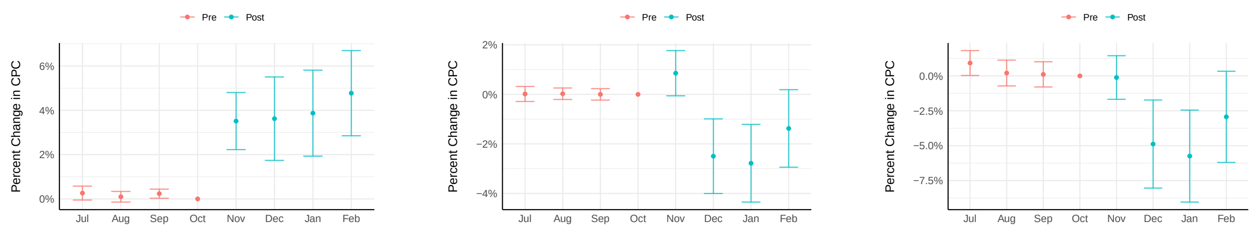
CPC results Our model predicts that CPC can increase or decrease depending on whether the query contains contextual information. In particular, CPC should increase for queries with limited to no contextual information; however, CPC may decrease for queries with contextual information.

Again, we group queries based on their length to operationalize and test these hypotheses. We then estimate Equation 5 and Equation 6 on the set of competitive queries, by query length, and using $log(CPC)$ as the dependent variable.[33] In Figure 10, we present the event

[33]We again remove non-competitive queries because, for these queries, prices can only increase since they

study estimates by query length. First, as before, in all cases, we find parallel pre-treatment trends, supporting the validity of our identification strategy. Second, we see that in the post-treatment period, CPC increases for short queries but decreases for medium and long queries.

Figure 10: Percent change in CPC by query length. Error bars represent 95% confidence intervals.



(a) Short queries. (b) Medium queries. (c) Long query.

In Tables 23, we present the estimates of Equation 5 by query length. Consistent with our model predictions, query length moderates the effect of BERT's introduction on existing auction market CPC. We find a 3.8% increase in CPC for short queries. However, as query length increases, CPC quickly decreases. CPC decreases by 1.5% for medium queries and 3.7% for long queries. These results suggest that BERT's introduction significantly enhances Google's ability to estimate more precise relevance scores for queries that contain contextual information, resulting in lower CPC but ultimately more relevant ads.

D.2 Year-over-Year DD Robustness Checks

We discuss three tests aimed at reinforcing our event study results. Specifically, we show that our results are robust to the inclusion of additional years as controls, changes in organic search ranking, and changes in consumer and advertiser search behavior. We find results consistent with our year-over-year DD. For the sake of brevity, we focus on replicating the effects for competitive queries, i.e., an increase in CS across all queries and heterogeneous

were zero when they were non-competitive.

Table 23: The effect of BERT on $log(CPC)$ by query length.

	(1) Short	(2) Medium	(3) Long
Post × Treated	0.038***	−0.015**	−0.038***
	(0.008)	(0.006)	(0.013)
Observations	47,912	88,888	23,887
R^2	0.686	0.728	0.671

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Estimated with competitive queries. Regressions include year-group, month, and query fixed effects. Standard errors clustered at the query level are reported in parentheses.

adjustments to CPC depending on the length of the query.

Additional years as controls Our year-over-year DD analysis uses query data from 2018 to 2020, with July 2018 to February 2019 as control units for July 2019 to February 2020. The downside of using only one year-month set (2018-2019) as a control window is we don't know whether it accurately represents the behavior of the auction market during those months. (This is equivalent to worrying about using only two states in a traditional state-level DD). As a robustness check, we re-estimate Equations 5 and 6 using 2016-2017 and 2017-2018 July-February months as additional control groups.

To identify competitive queries with this additional data, we take the mode Competition Score across all July–October months and years (2016–2019, generating 16 observations per query) and deem a query non-competitive if the mode is 0.

We repeat our regressions by query length for $log(CS)$ and $log(CPC)$. We report these results in Tables 24 and 25. The results are directionally consistent and similar in magnitude to those estimated in Tables 22 and 23. The fact that using additional control years leads to broadly similar results suggests we are reasonably controlling for within-unit seasonal patterns and that the 2018-2019 auction market outcomes are fairly representative of the average Google market in the absence of BERT.

Table 24: The effect of BERT on $log(CS)$ by query length.

	(1) Short	(2) Medium	(3) Long
Post × Treated	0.012***	0.009***	0.017***
	(0.001)	(0.001)	(0.001)
Observations	83,765	154,872	36,927
R^2	0.917	0.956	0.905

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Estimated with competitive queries. Regressions include year-group, month, and query fixed effects. Standard errors clustered at the query level are reported in parentheses. Data from 2016 to 2020.

Changes in organic rank results To address concerns regarding shifts in organic rankings, we reuse the collected monthly organic rankings data from SEMRush for the queries in our data. Using our DD identification strategy, we test for changes in these front-end organic rank changes. We report these results in Table 26. We find that there is a roughly 11.5% increase in the likelihood a query contains a featured snippet, the number of SERP features is generally increasing by about 3.7%, and there is a 4.3% increase in the likelihood that the top domain link changes after BERT. Google said that BERT would meaningfully change the result pages of about 10% of queries and increase the presence of featured snippets. Our estimates seem to capture changes in organic search rankings that are consistent with Google's statement.

To test for this possibility, we re-estimate Equation 5 but controlling for organic result changes. We present these results in Tables 27 and 28 for CS and CPC, respectively.

We find that the CS estimates are almost the same as those reported in Table 22. For CPC, we find that controlling for organic rank variables slightly attenuates the estimates reported in Table 23. These results suggest that BERT likely did affect organic rank changes, but these changes don't appear to be the primary mechanism driving our empirical results.

Table 25: The effect of BERT on $log(CPC)$ by query length.

	(1) Short	(2) Medium	(3) Long
Post × Treated	0.032***	−0.021***	−0.058***
	(0.009)	(0.006)	(0.013)
Observations	83,765	154,872	36,927
R^2	0.629	0.702	0.627

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Estimated with competitive queries. Regressions include year-group, month, and query fixed effects. Standard errors clustered at the query level are reported in parentheses. Data from 2016 to 2020.

Table 26: Changes in organic results.

	(1) Feature Snippet	(2) log(SERP Features Count)	(3) Domain Change
Post × Treated	0.115***	0.037***	0.043***
	(0.004)	(0.004)	(0.005)
Observations	104,228	104,228	104,176
R^2	0.565	0.655	0.207

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Estimated with competitive queries. Regressions include year-group, month, and query fixed effects. Standard errors clustered at the query level are reported in parentheses.

Changes in consumer behavior Another potential explanation is that BERT affects consumer search behavior, which, in turn, affects CPC and competition. For example, consider the case in which consumers learn that results are getting better for longer/complex queries because of BERT, and this changes the types of queries consumers are searching for. Advertises, then, may start targeting and bidding for these newly popular queries, which can lead to changes in competition and CPC.

We perform a test to partially address this concern. We collect data on Google Trends

Table 27: Changes in $log(CS)$ by query length, controlling for organic rank changes.

	(1) Short	(2) Medium	(3) Long
Post × Treated	0.013***	0.009***	0.014***
	(0.001)	(0.001)	(0.002)
Feature Snippet	0.001	−0.001	0.005***
	(0.003)	(0.002)	(0.002)
Top Domain Change	0.001	−0.00003	0.001
	(0.001)	(0.001)	(0.001)
SERP Features Count	0.0001	0.0002	−0.002**
	(0.001)	(0.0005)	(0.001)
Observations	30,546	57,048	16,582
R^2	0.949	0.969	0.934

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Estimated with competitive queries. Regressions include year-group, month, and query fixed effects. Standard errors clustered at the query level are reported in parentheses.

search interest and search volume from SEMRush for each query in our dataset. We then use these variables to account for each query's popularity. We re-estimate Equation 5, including these variables as control. We report these results in Tables 29 and 30. We find that controlling for Google trends and estimated search volumes does not affect the estimates.

Table 28: Changes in $log(CPC)$ by query length, controlling for organic rank changes.

	(1) Short	(2) Medium	(3) Long
Post × Treated	0.024^{**}	-0.018^{**}	-0.030^{**}
	(0.010)	(0.008)	(0.015)
Feature Snippet	-0.008	-0.018^{*}	-0.017
	(0.015)	(0.010)	(0.017)
Domain Change	-0.014^{**}	0.005	-0.009
	(0.006)	(0.005)	(0.010)
SERP Features Count	0.014^{***}	0.008^{**}	0.029^{***}
	(0.004)	(0.003)	(0.007)
Observations	30,546	57,048	16,582
R^2	0.696	0.736	0.690

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Estimated with competitive queries. Regressions include year-group, month, and query fixed effects. Standard errors clustered at the query level are reported in parentheses.

Table 29: The effect of BERT on $log(CS)$ by query length controlling for search interest and search volume.

	(1) Short	(2) Medium	(3) Long
Post × Treated	0.012***	0.010***	0.013***
	(0.001)	(0.001)	(0.002)
Interest	0.0001	0.00002	0.0001
	(0.0001)	(0.00005)	(0.0001)
log(SV)	0.011***	0.003	0.003
	(0.004)	(0.003)	(0.003)
Observations	47,864	88,840	23,807
R^2	0.951	0.970	0.919

Significance Levels: *p<0.1; **p<0.05; ***p<0.01.
Note: Estimated with competitive queries. Regressions include year-group, month, and query fixed effects. Standard errors clustered at the query level are reported in parentheses.

Table 30: The effect of BERT on $log(CPC)$ by query length controlling for search interest and search volume.

	(1) Short	(2) Medium	(3) Long
Post × Treated	0.032^{***}	-0.017^{***}	-0.041^{***}
	(0.008)	(0.006)	(0.013)
Interest	-0.0002	0.0001	0.0005
	(0.0003)	(0.0003)	(0.0005)
log(SV)	0.065^{***}	0.041^{***}	0.102^{***}
	(0.013)	(0.009)	(0.018)
Observations	47,864	88,840	23,807
R^2	0.686	0.728	0.670

Significance Levels: $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$.

Note: Estimated with competitive queries. Regressions include year-group, month, and query fixed effects. Standard errors clustered at the query level are reported in parentheses.